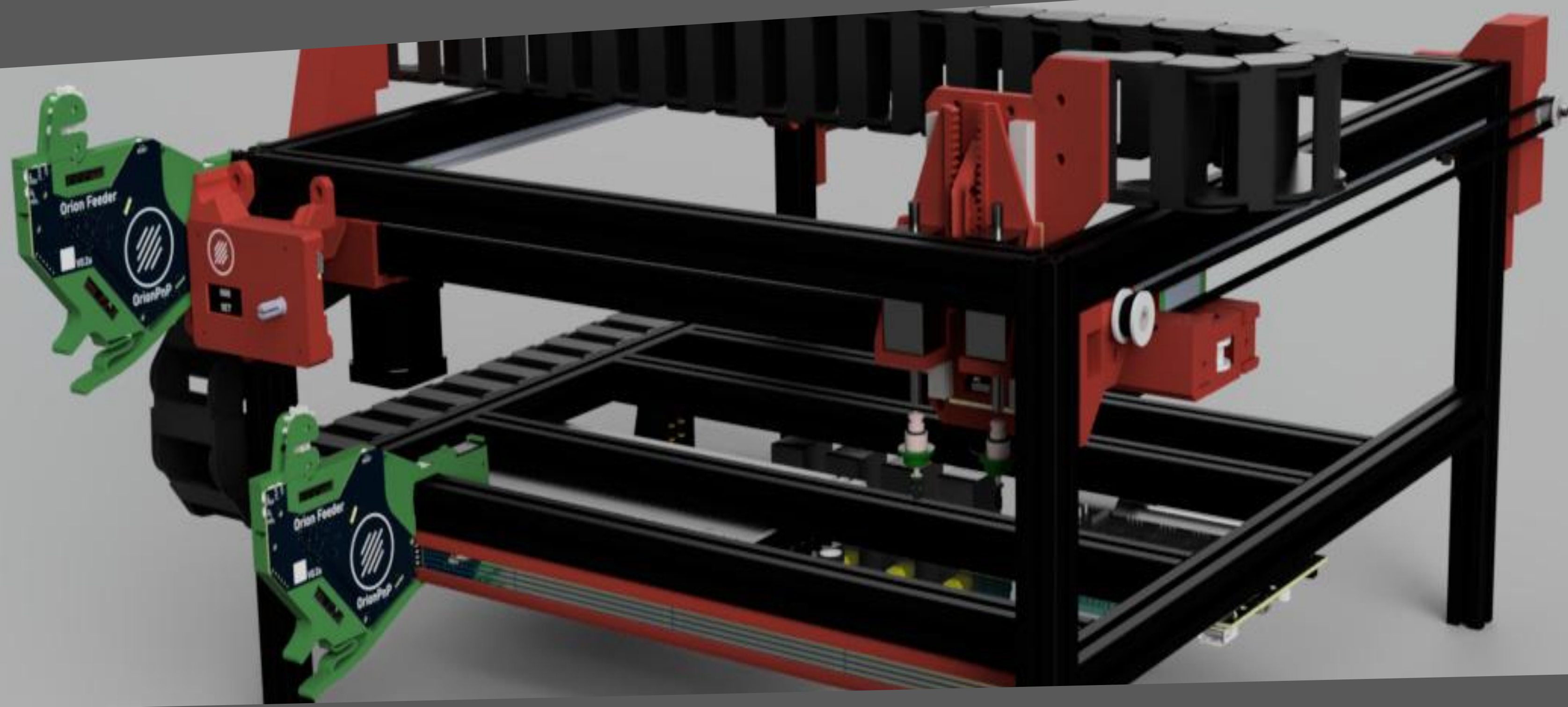
OrionPnP

Assembly manual



Rev.
01a

Before we start…

A word about safety

You are about to build a machine, powered by mains voltage, which could become DEADLY if not handled properly. Please consult a professional if you are unsure on how to proceed when the symbol is shown.

We are going to build a machine, capable of injuring you and/or others. There is no pride in getting hurt or worse, so be careful and always wear adequate protective gear when needed.

In the manual the symbol is used whenever special care is required. Make sure to read the manual before starting!

A word about this project

OrionPnP

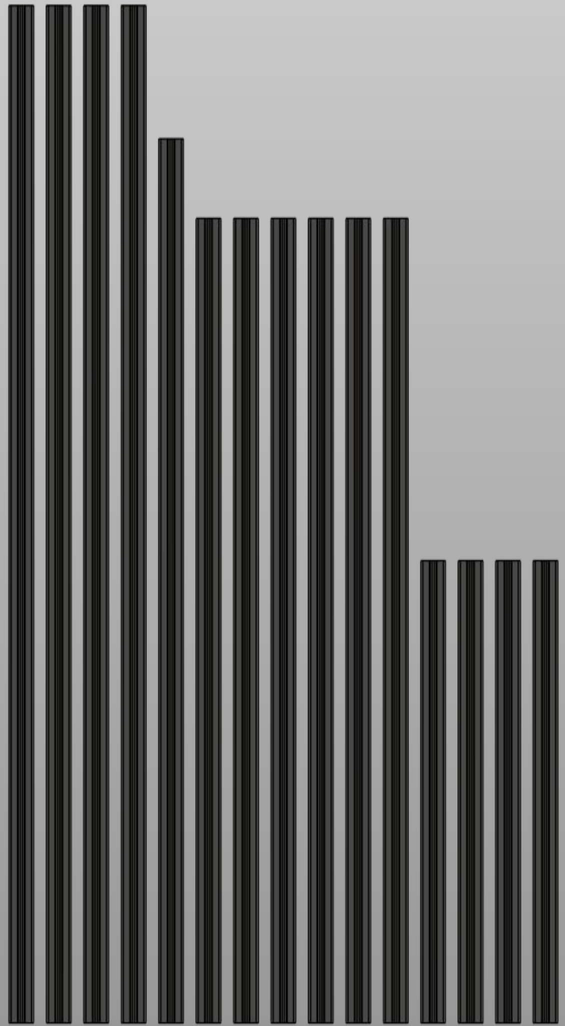
The OrionPnP project has been conceived to bring good performance pick-and-place machines to makers, with solid and reliable operation, the project aims to handle 0402 components to empower makers and allow smaller boards to be manufactured. The heart of the project is the open source software OpenPNP of course!
The project is open source, and as such every reference in this manual is made to a specific machine version. Mods, add-ons and other non official modifications will have their own guides, so follow those.
The source of each and every information is (and will always be) the Github repository.
Go and check the project if you haven't already, read the wiki and if this is the project for you, get ready to build!

How to navigate this manual

The manual

This manual will take you step by step from a bunch of components to a working OrionPnP machine. Each step is written as clearly as possible, but the drawings and the representations should clarify any doubt.
Minor differences might be noticeable but if the difference doesn’t affect the assembly, the step is still valid. *(e.g. toothed profile on belt holders might be different)*
Several tools will be needed, and the list can be found in the BOM. 3D printed parts require no particular setting and can be printed in PLA, only motor mounts should be printed in heat resistant material as ABS, ASA or PETG. If using PLA for all, print the motor spacers in ABS or ASA to prevent issues!
Several screws will be used, make sure to reference the BOM for the codes relative to those. A 1:1 scale page of most screws and fasteners is in the works. For this to be useful you must print it without scaling from the printer!

Main frame



Get the profiles ready

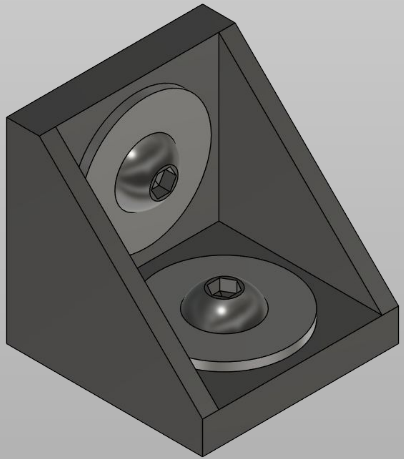
The machine is built around 2020 aluminium profiles, their sizes and amounts are:

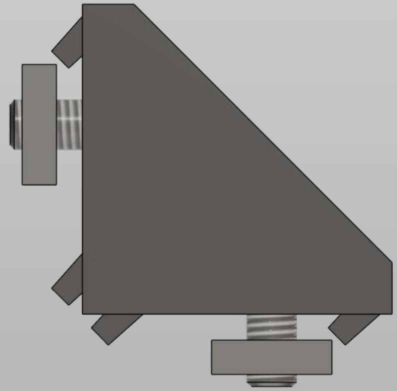
4x 250mm for the legs	6x 435mm for the X axis and bracing
1x 478mm for the X axis itself	4x 550mm for the Y axis

All these profiles need to be prepared for the next step.
No drilling or tapping is required by design, but you can tap the ends and use screws to reinforce the connections if you *really* want to.
We will start the build by assembling the "bed", where the work surface will go later on and the rest of the frame.

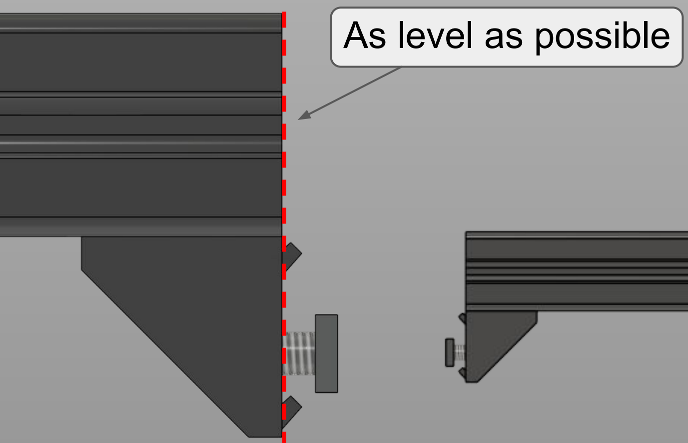
Main frame

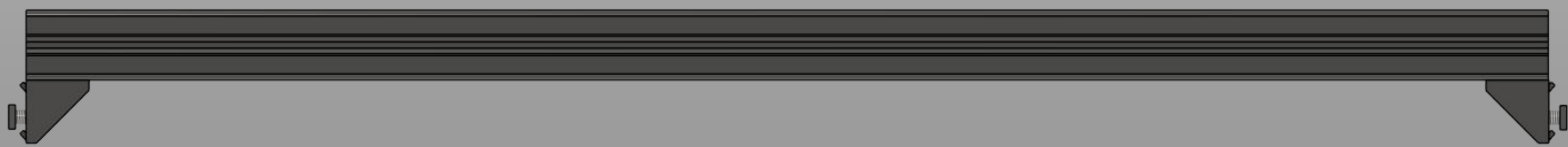
Prepare corner brackets

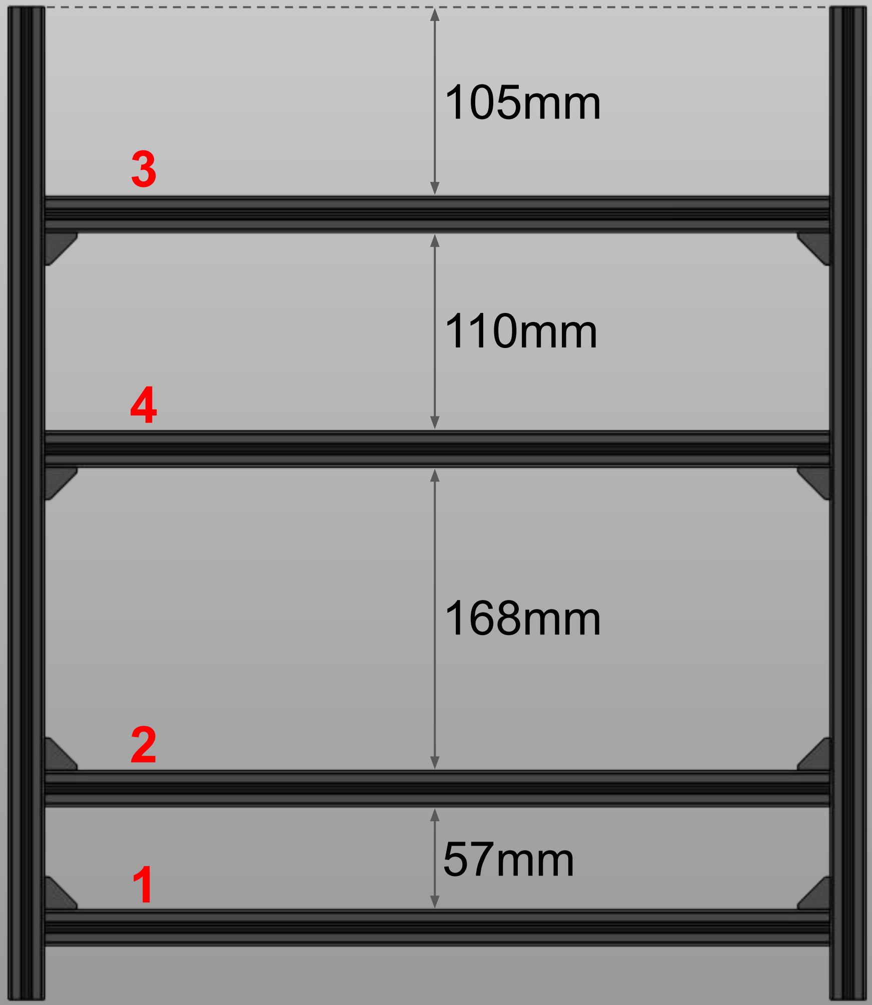




Prepare the corner brackets as illustrated, using M4x8mm button head screws and a square nut. A washer is recommended.
In total we need 20 brackets prepared this way.
Note: depending on your corner brackets you might also use M5x8mm screws.
Then prepare the 6x 435mm profiles as illustrated below. Do the same for 2x 550mm profiles. Only tighten screws on the profile side of the bracket. Leave the other nut loose!







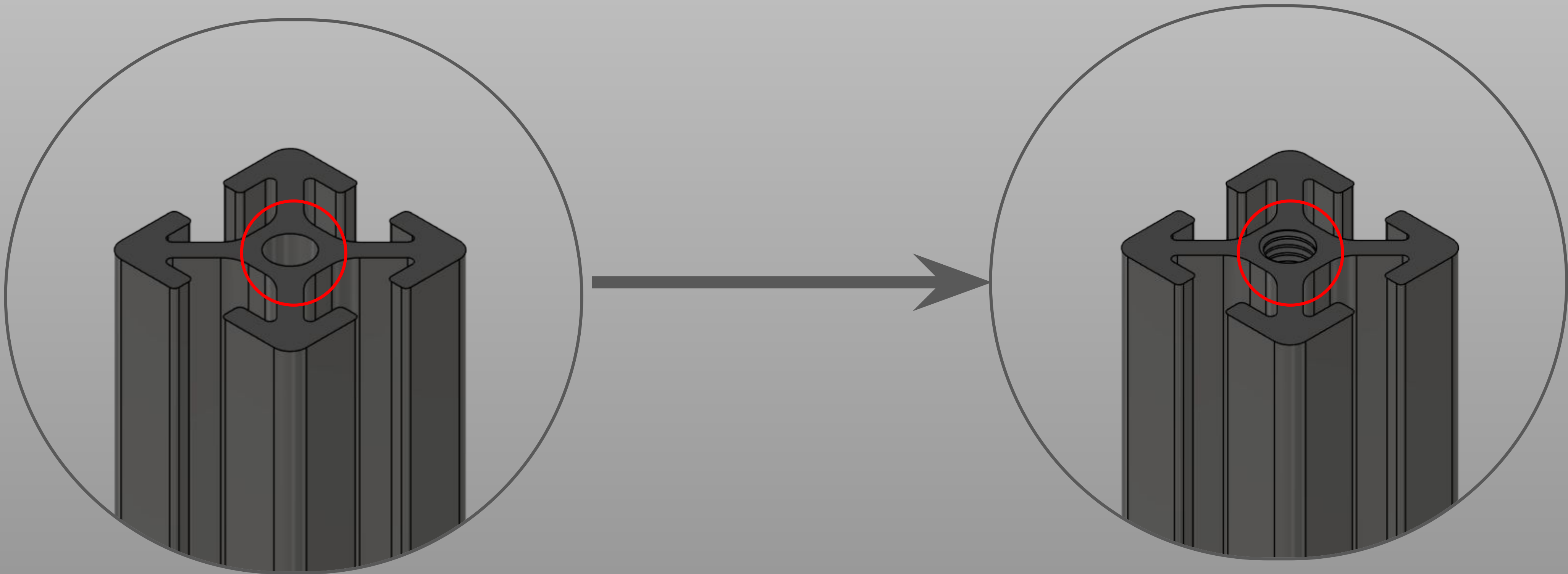
Assemble the bottom profiles

Take 4x of the 435mm profiles prepared earlier and 2x 550mm profiles.
Place them as shown in the drawing, following the order, making sure the measurements are as accurate as possible.
It is not mandatory to be 0.1mm perfect, we will calibrate in software, but try to be as precise as possible.
Pay attention to the orientation of the brackets!

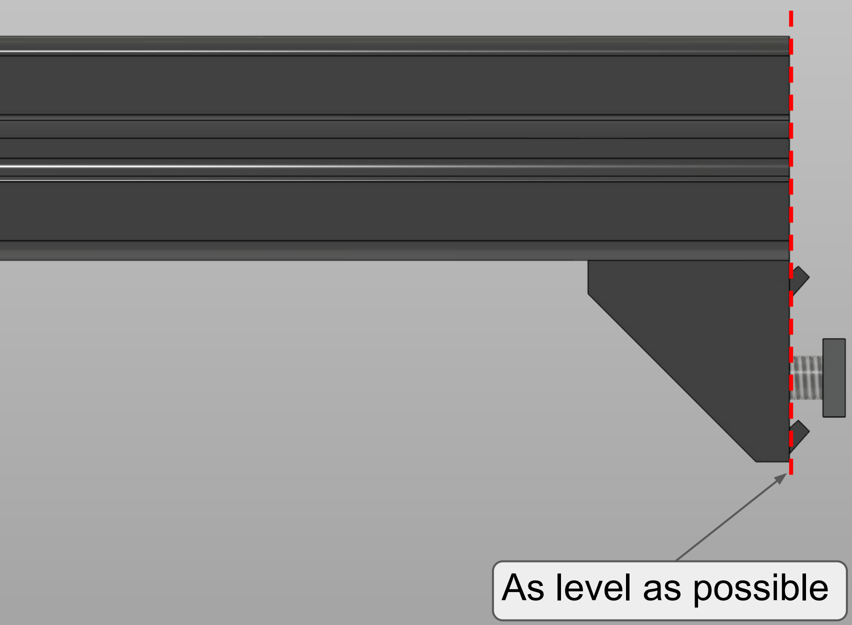
Note: *illustration is on scale so yours should look very similar to this.*

Tap the legs of the frame

Take the 4 250mm profiles and tap one end with an M5 thread. This will be used to fix the feet on them later on. Only about 6mm need to be threaded.



Prepare top Y beams brackets



Take the other 2x 550mm profiles and add corner brackets as shown in the picture.
Tighten only the section connected to the profile.
Then keep them to the side as we will need them later.

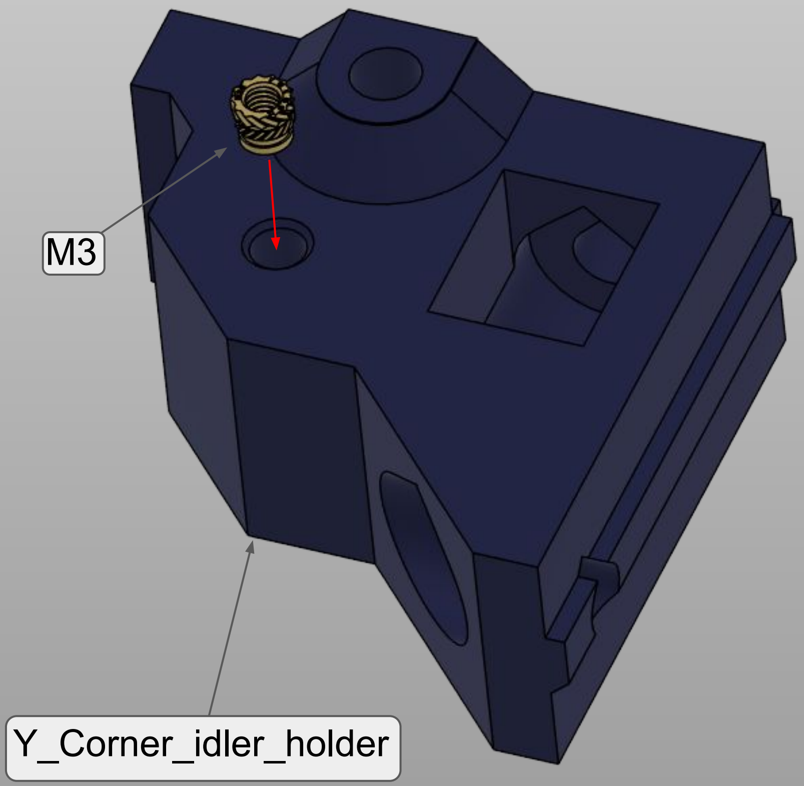
Now we will move on the first two 3D printed parts.

Prepare the idler holders

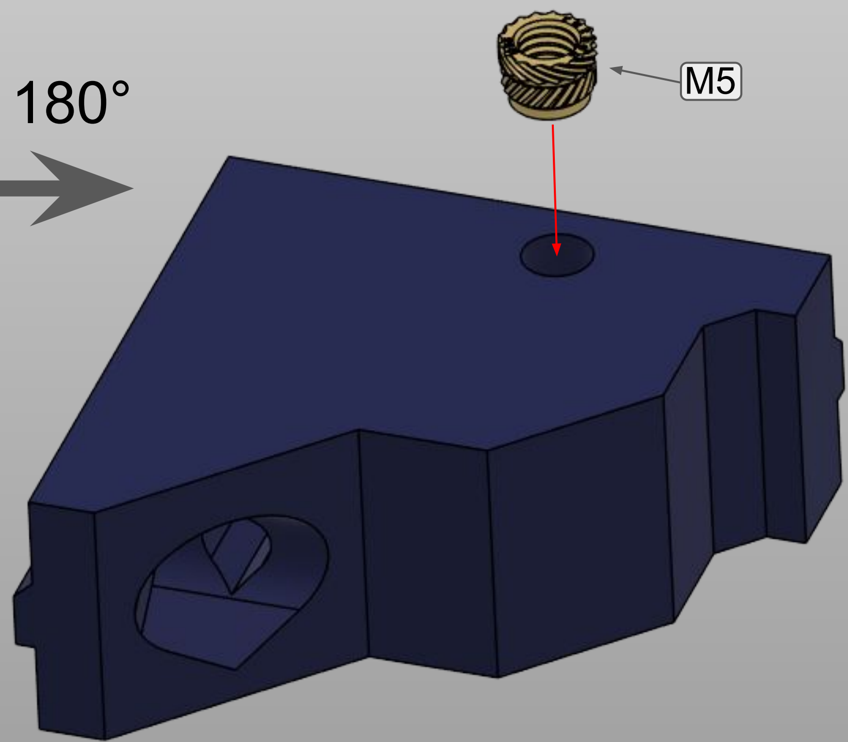
We will need the *Y_Corner_idler_holder_R* and *Y_Corner_idler_holder_L* files printed.
We will also need:
2x M3 heat set inserts
2x M5 heat set inserts
4x M5 square nuts
4x M5x10mm button head screws
2x M5x35mm cap head screws
2x Toothed GT2 idlers for 6mm belts

Steps will need to be done on both R and L pieces, they are symmetrical.

Main frame

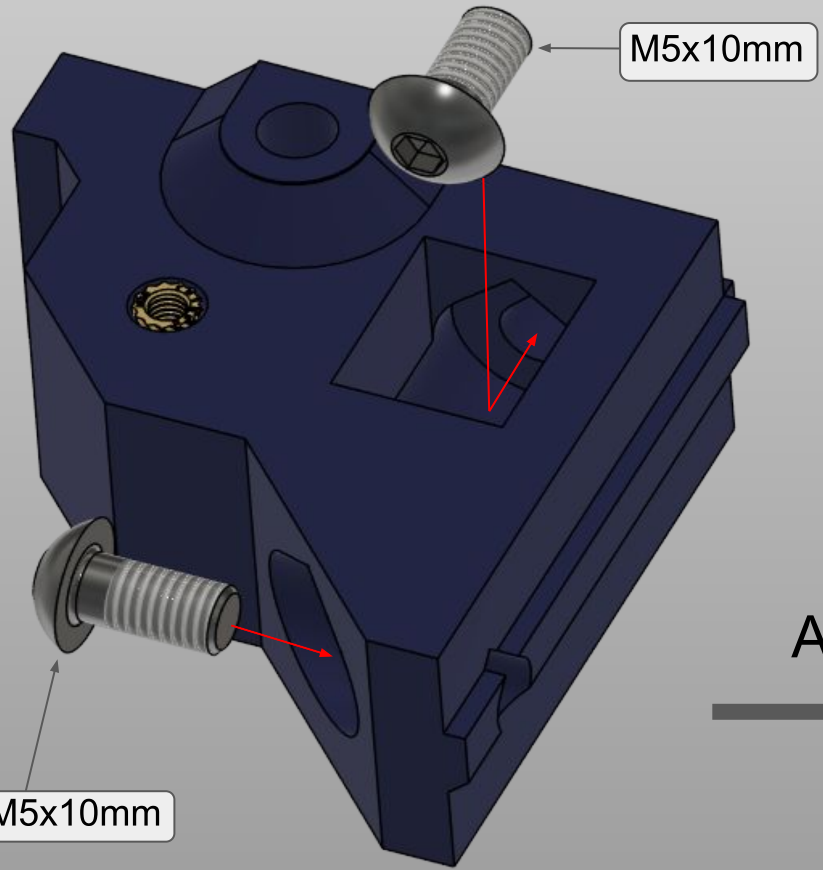


Turn the piece 180°



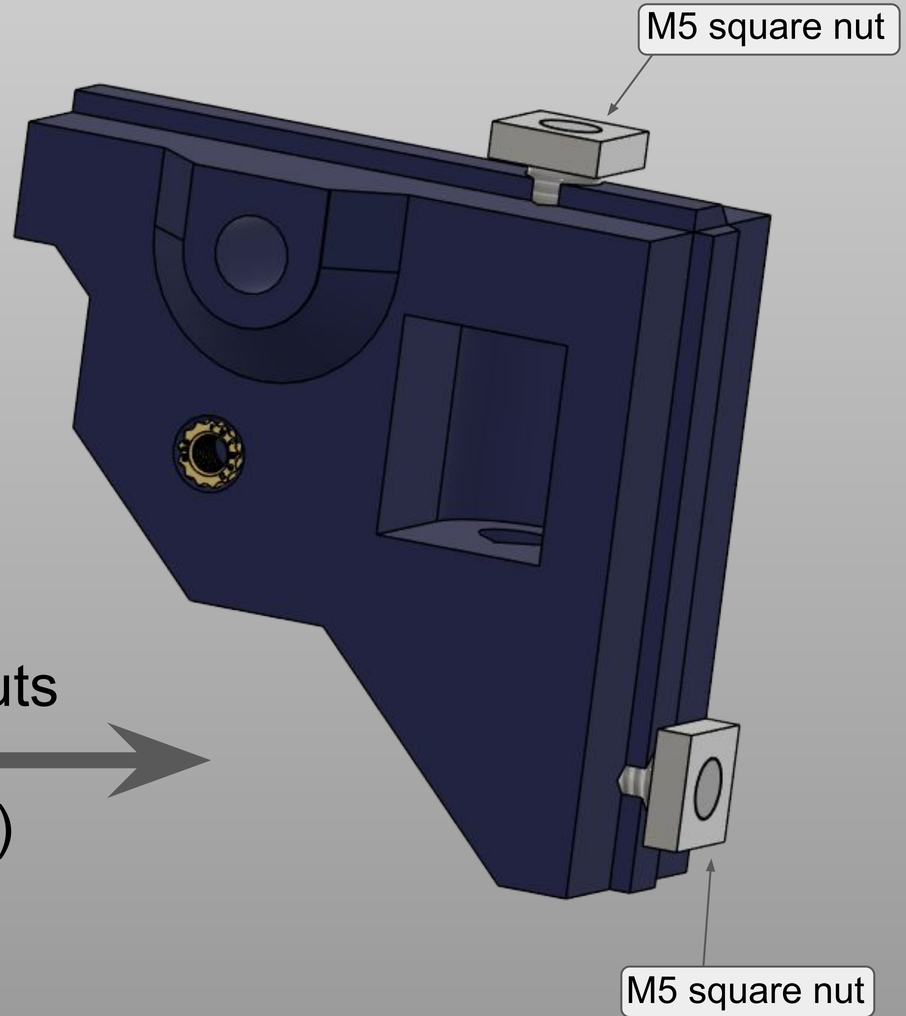
Note: *L idler holder shown in the pictures*

Main frame



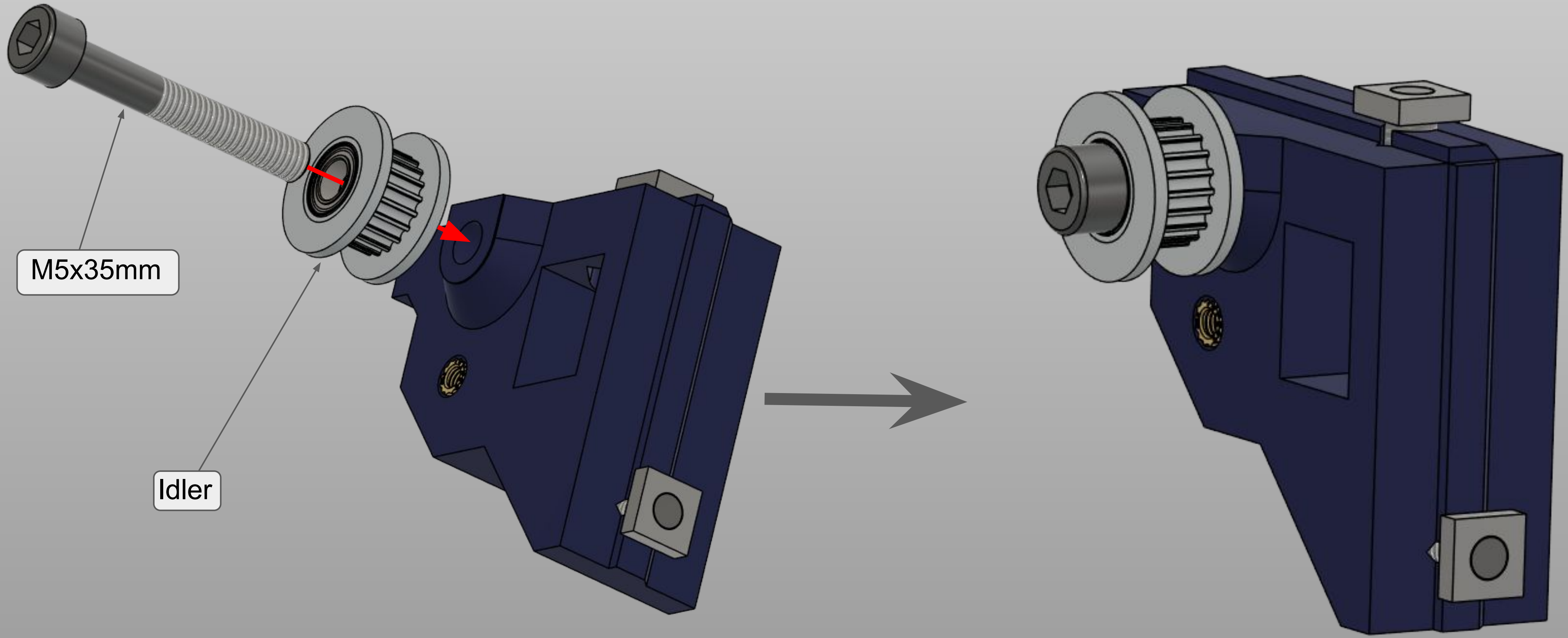
Add square nuts

(leave loose)



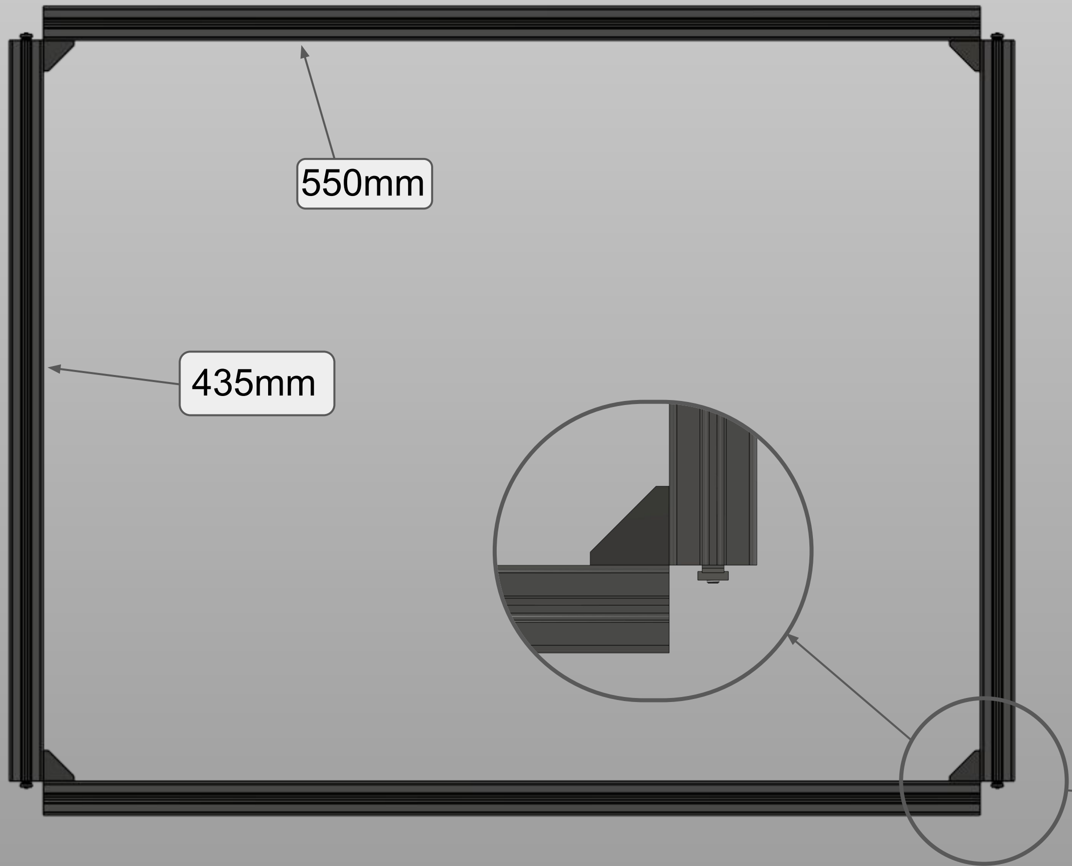
Note: *L idler holder shown in the pictures*

Main frame



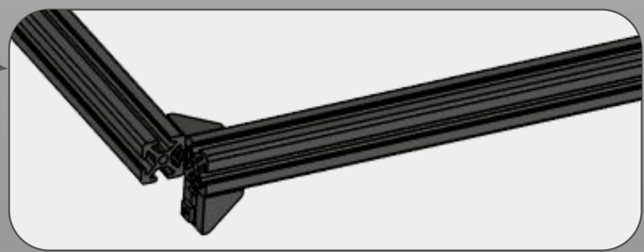
Note: *L idler holder shown in the pictures*

Main frame

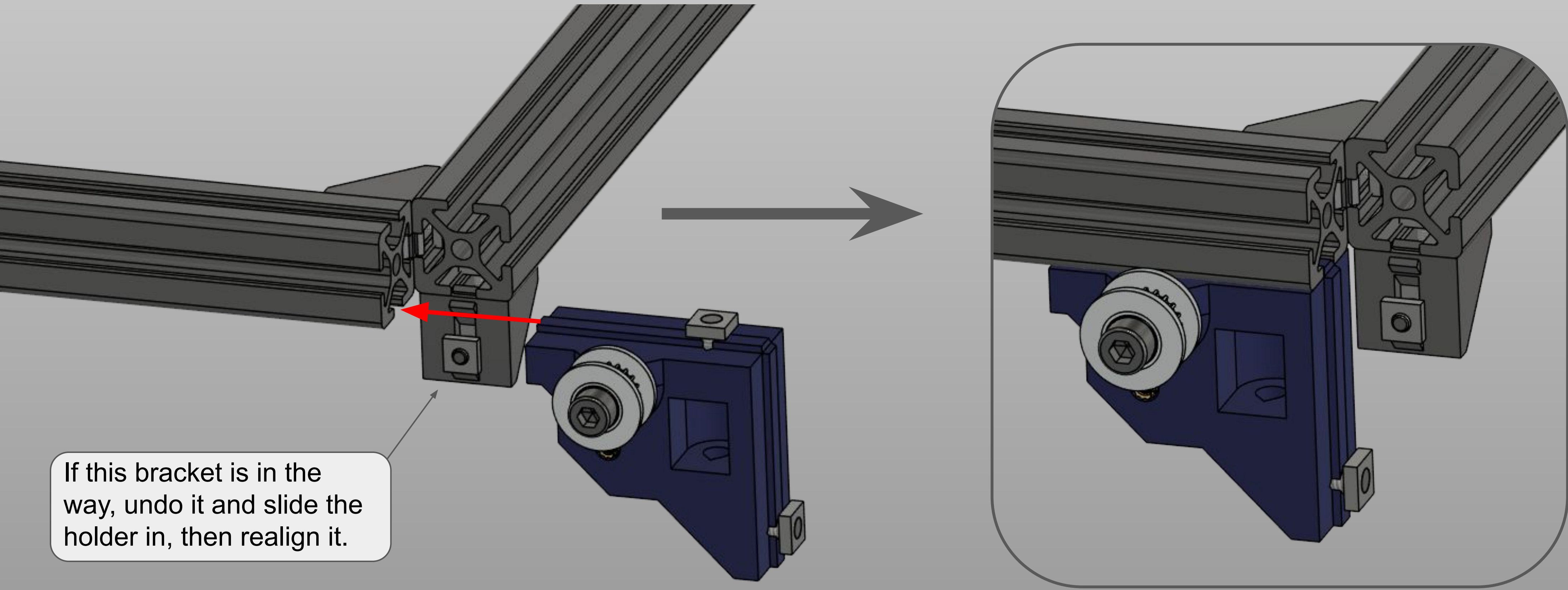


Assemble the top frame

Now take the 2x 550mm and 2x 435mm profiles prepared earlier and assemble as shown. Be careful of the orientations, and make sure the two profiles on the edge are even like shown.
Later the leg profiles will need to fit here.

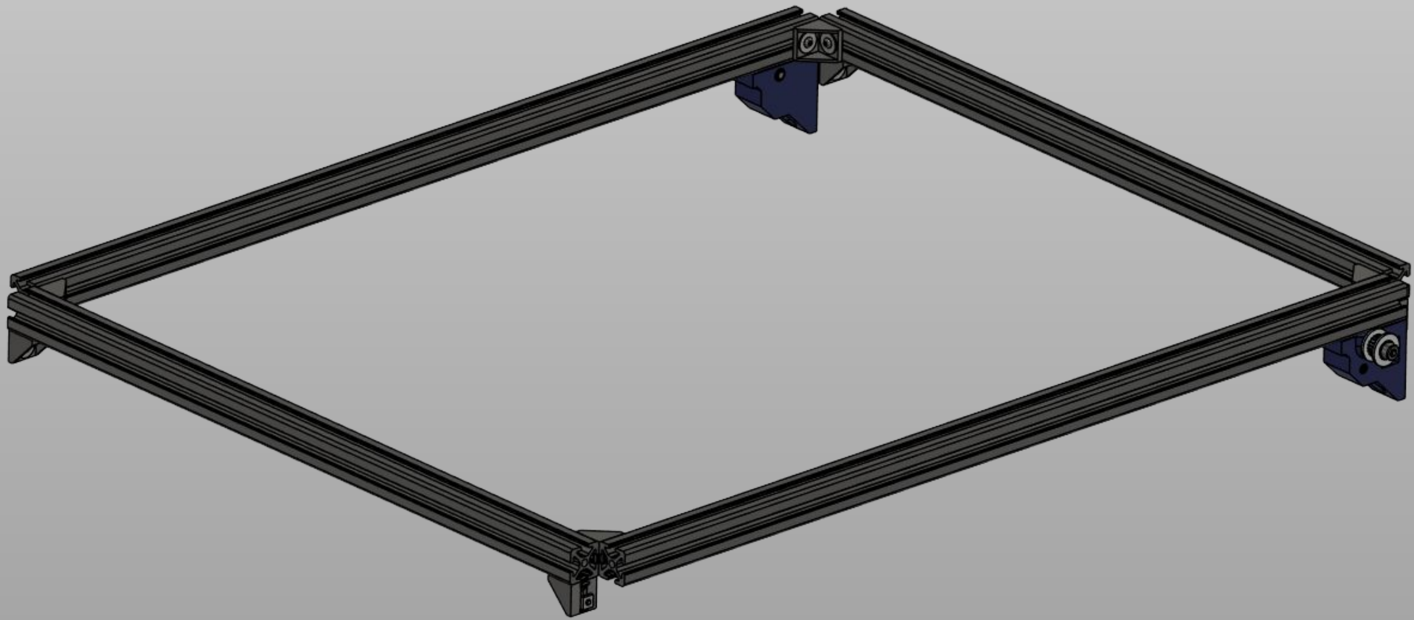


Mount the idler holders



Note: *L idler holder shown in the pictures*

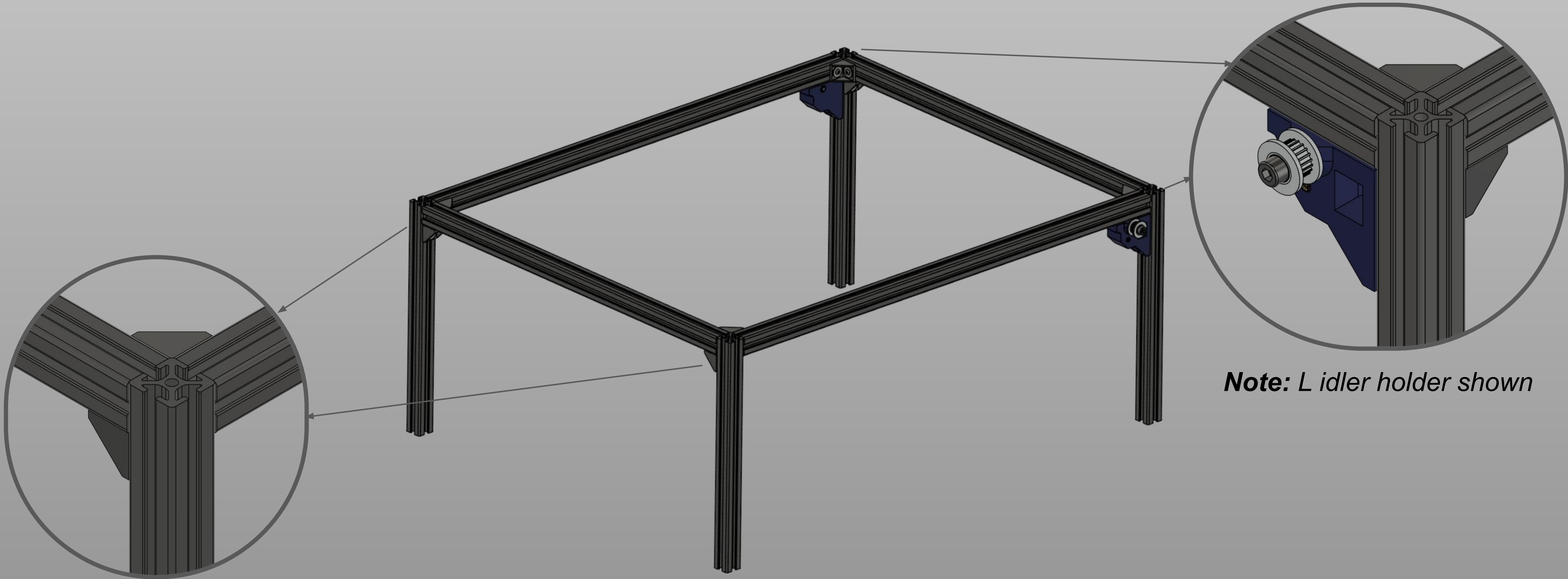
Finished top frame



The top frame is now finished, make sure it looks like the picture shown.
Do not worry at this point if it is not aligned, we will take care in the next steps about it.

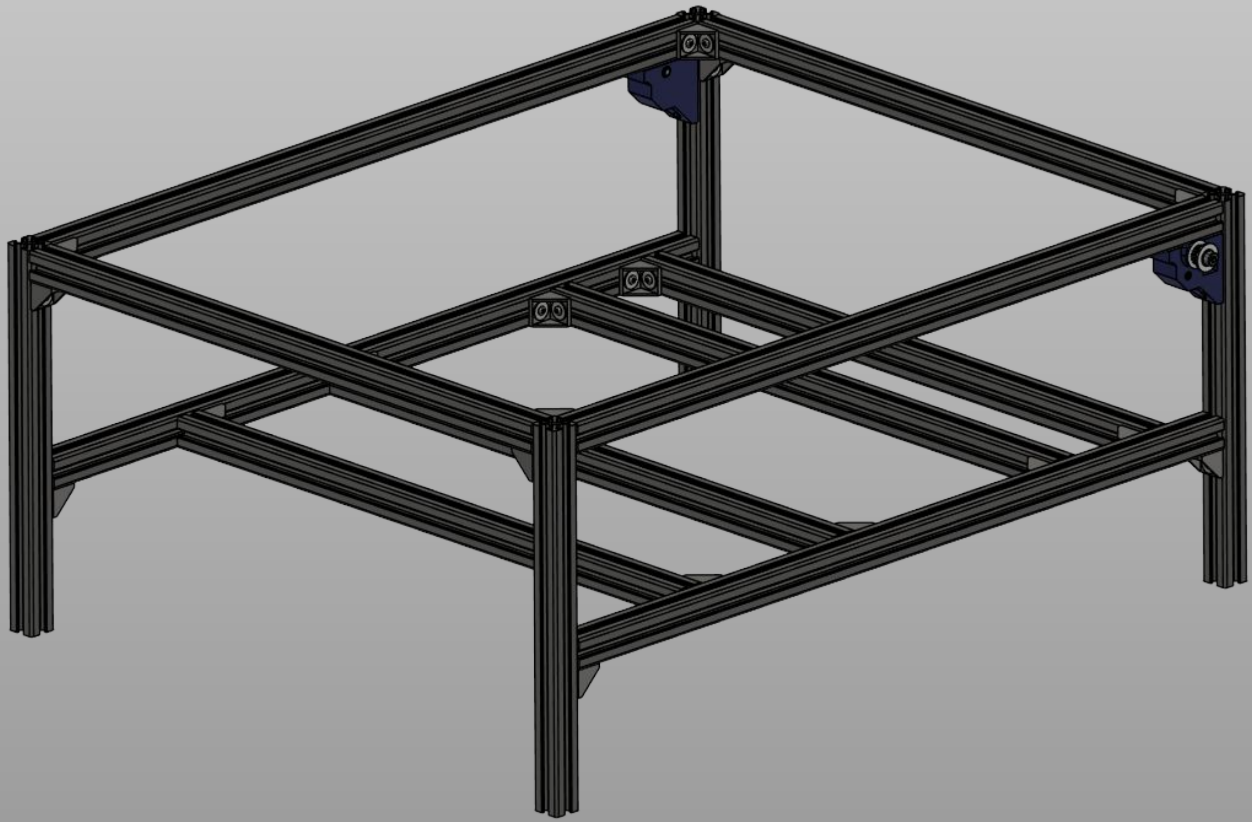
Finished top frame

Take the 4x 250mm profiles and fix them to the top frame as shown.

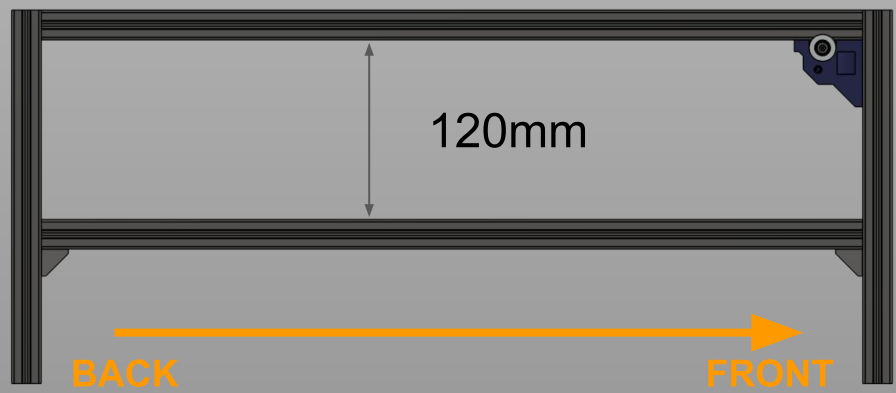


Note: *L idler holder shown*

Finish the frame



Now slide in the bottom frame as shown and screw the corner brackets to fix it, take care about the measurements referred below.

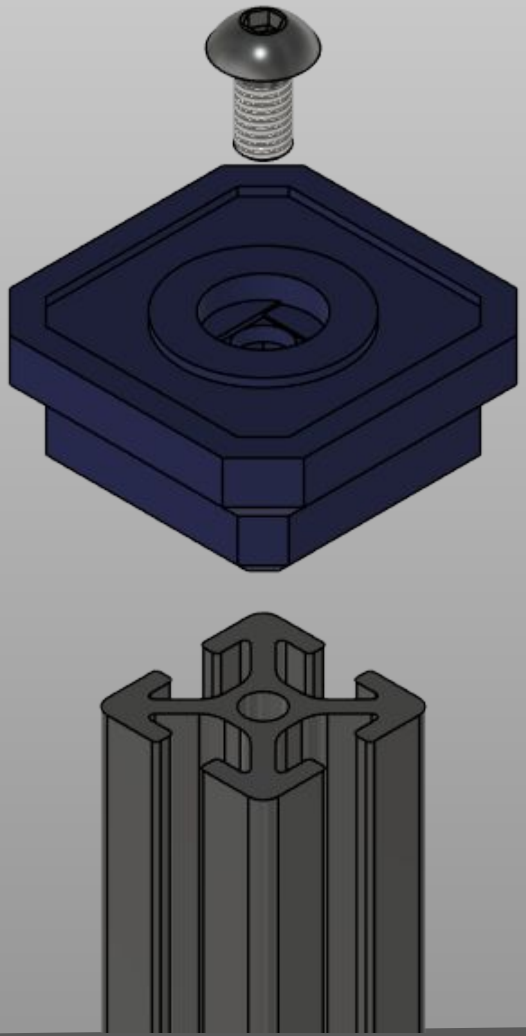


***Note:** Keep in mind the front placement!*

Square the frame

This is the time to square the frame. To do this lay the machine on the top frame, having care to check if there is any wobble.
Undo screws and fix all the profiles in their correct positions, making sure to measure the distances when needed.
If the frame is square, you should have all diagonals being equal (or close enough).
Once you are done, we can move on to the next section: rails!

Attach feet



Depending on the revision of the CAD, you might have different feet designs, but they all fix to the machine in the same way. In the pictures we will show the standard feet. They would require a rubberized insert to add grip.
There is a “tilt” foot for the back of the machine available, this allows to easily tilt the OrionPnP for working easier on its electronics etc…

Y Axis rails

Prepare the rails and fasteners

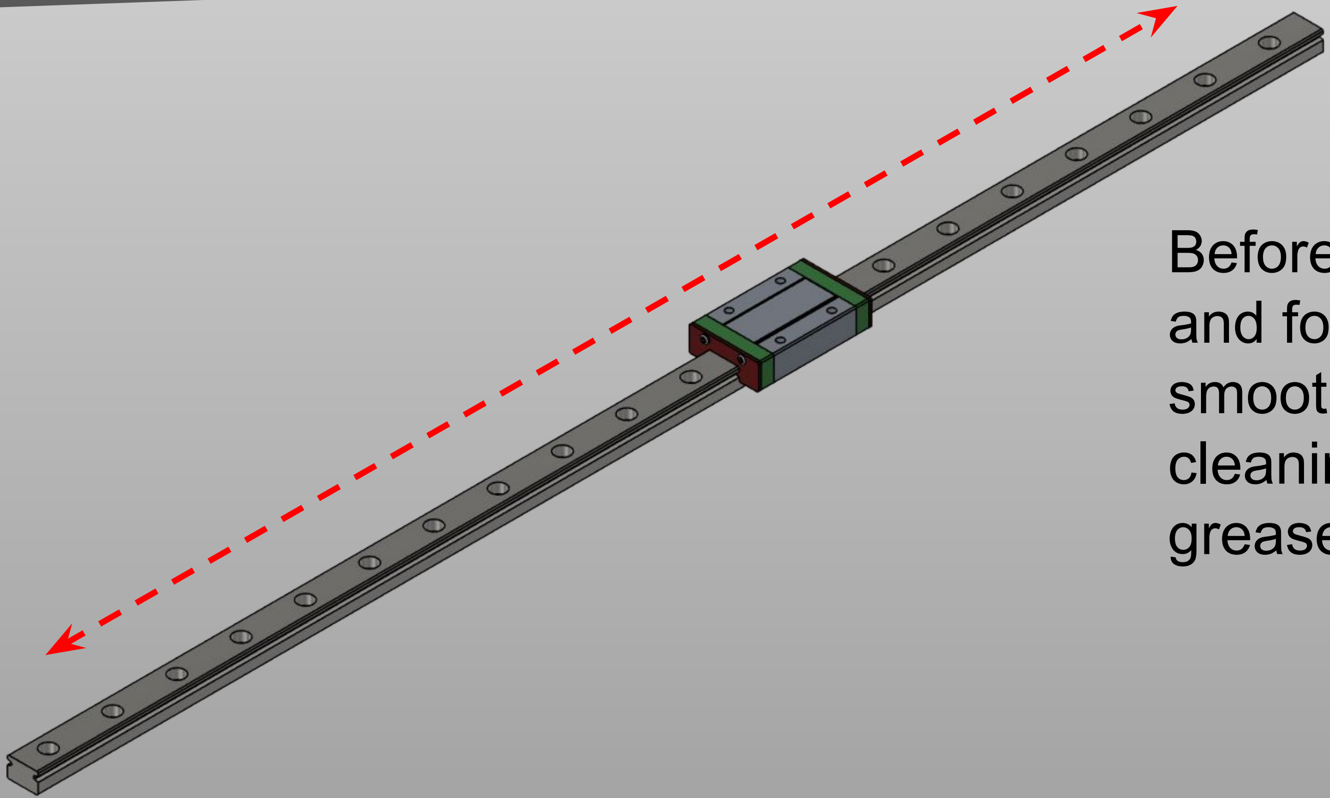
For this section, we will need 2x 500mm MGN12 linear rails to make the two Y axis rails. We need to clean them with some isopropyl alcohol to degrease them first! Remove the shuttles if you have the right holder for the carriage, otherwise leave it on!

Take care while removing the carriages to not losing the balls inside! There are plenty of tutorials on how to properly clean rails!

When clean, regrease it properly and get ready these fasteners:

- 20x M3x8mm Socket head screws
- 20x M3 hammer T-nut

Y Axis rails

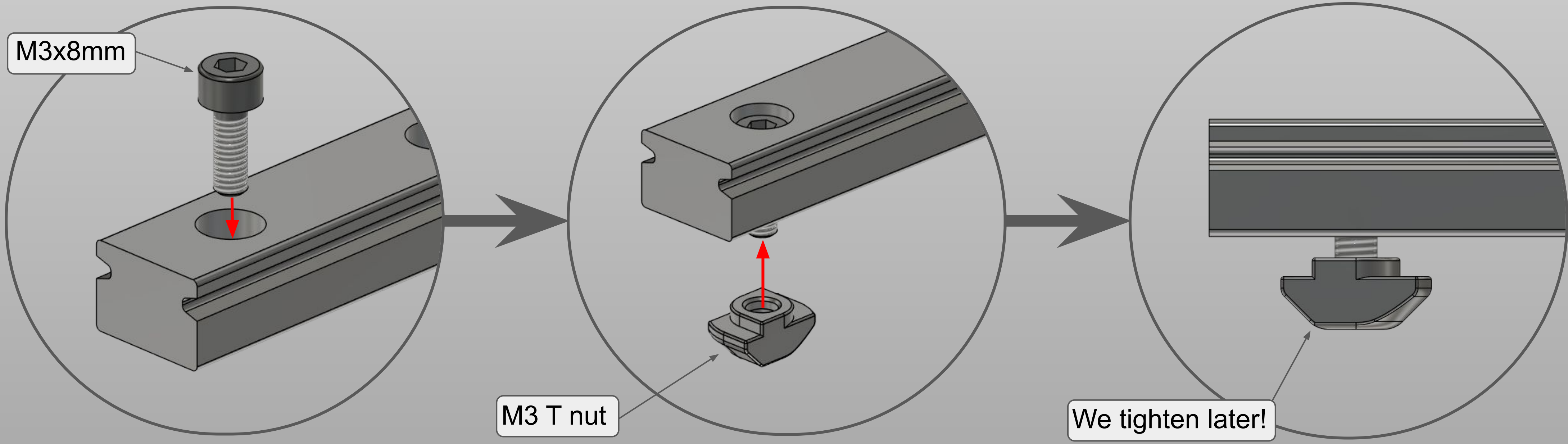


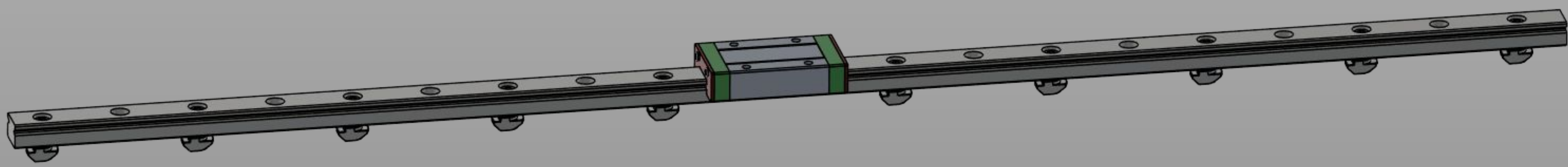
Check the rails

Before moving on, slide the rails back and forth as shown. Check if you can smoothly move them, if not try cleaning them better and applying grease if you did not earlier.

NEVER LET THE CARRIAGE SLIDE OFF!

Y Axis rails

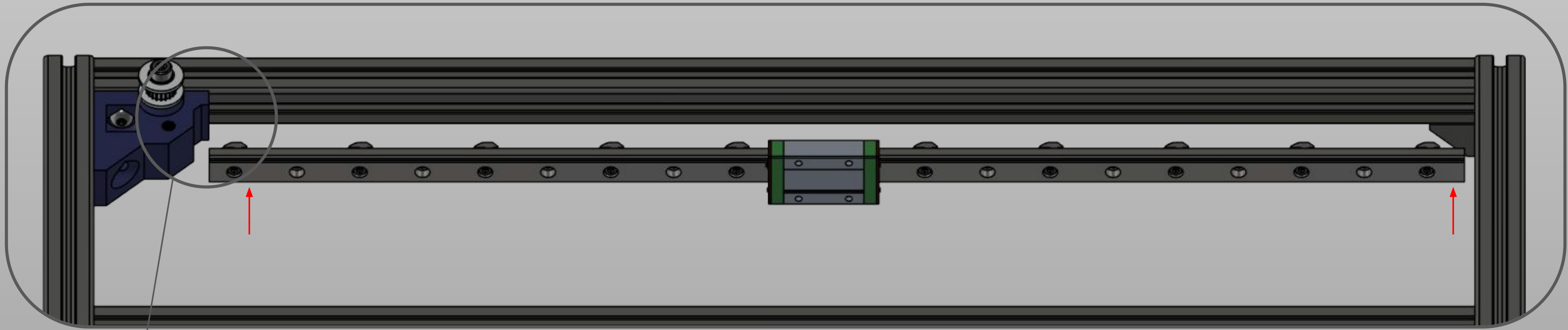


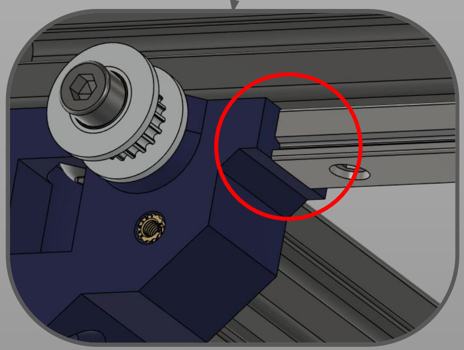


Note: *We need just 10x screws per rail. You can add more but they are not necessary. Space them as shown.*

Y Axis rails

Mount Y rails



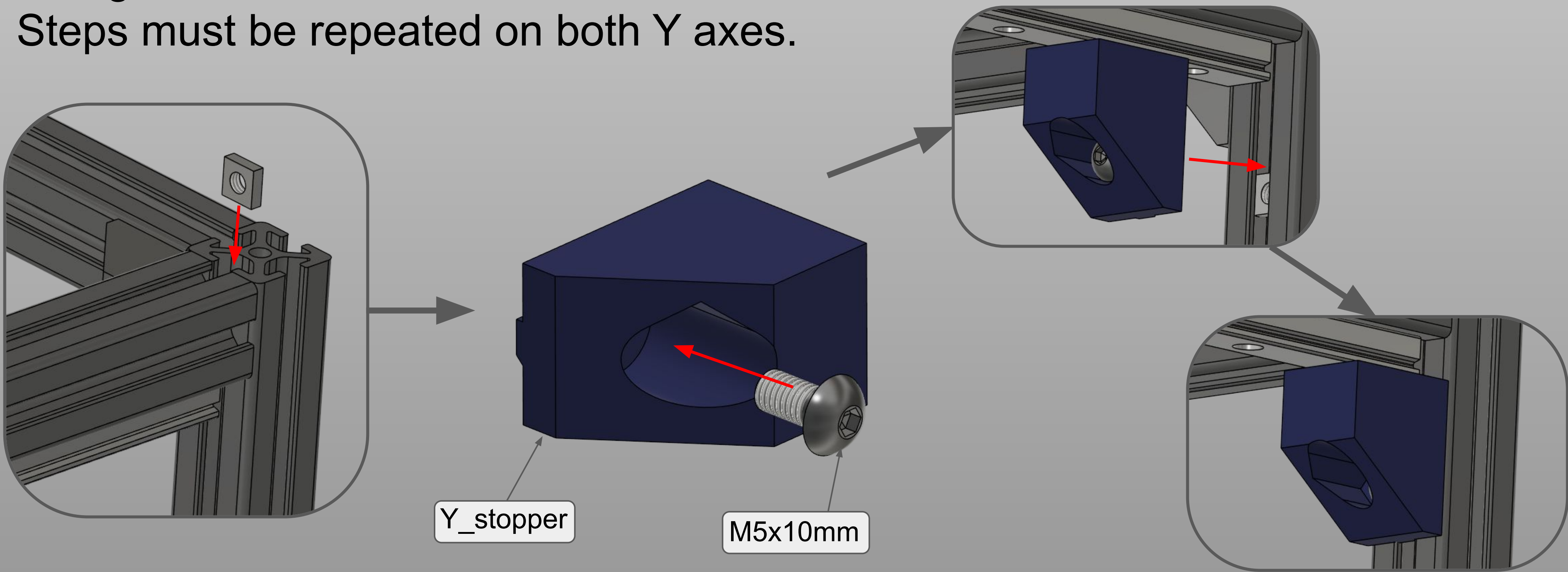


Now mount the rails, making sure to tighten gradually the screws, starting from one side to keep it even.
Pay attention, the rail should bump into the idler holders!

Repeat for both rails before moving on!

Mount Y stoppers

We will mount the two Y stoppers now, needed to prevent the carriage from falling out and to act as blocks for the Y axis.
Steps must be repeated on both Y axes.



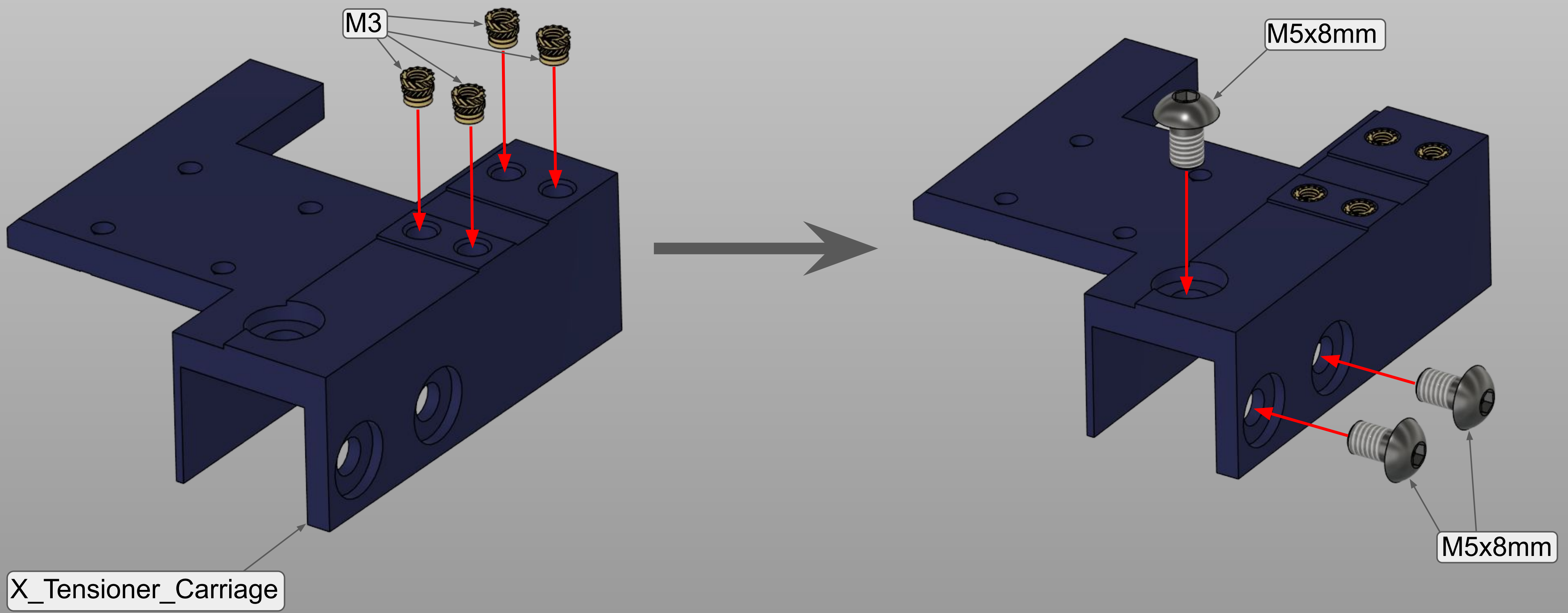
Prepare the rails and printed parts

For the X axis we will need the 400mm MGN12 rail, once again we need to clean it from the oil and grease it as before:

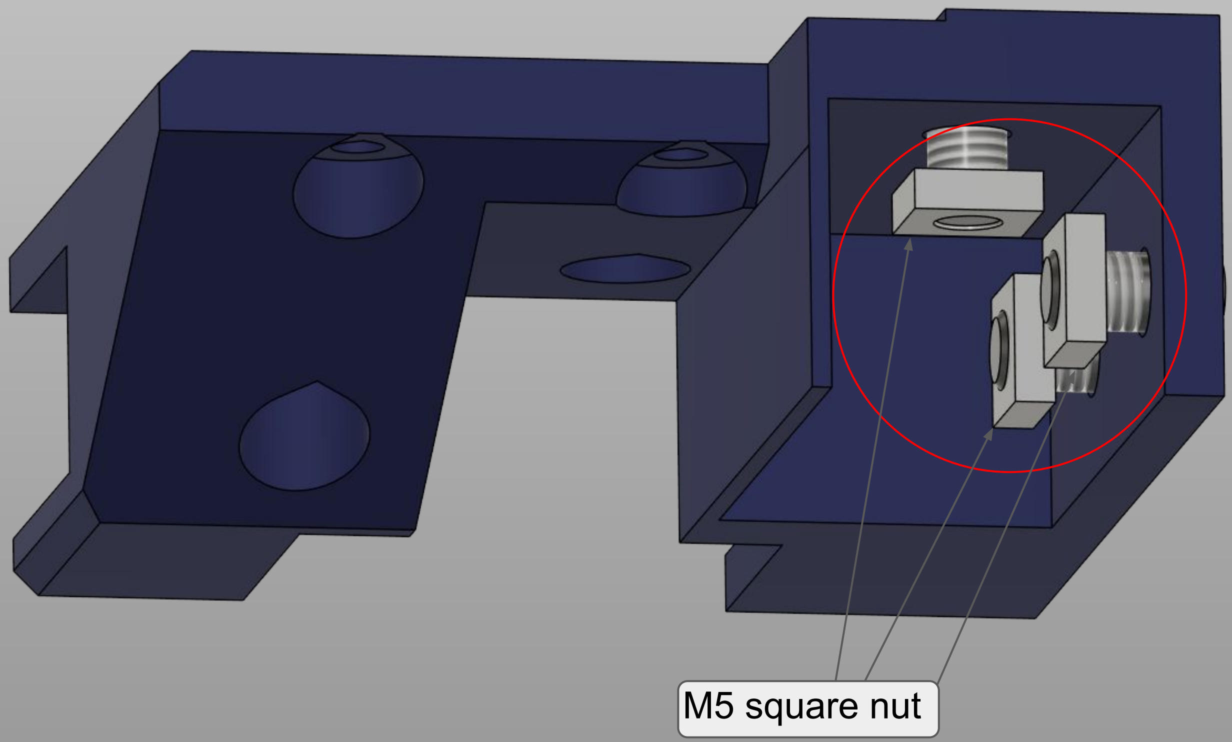
> Take care while removing the carriages to not losing the balls inside! There are plenty of tutorials on how to properly clean rails!

When clean, regrease it properly and get ready to assemble. We will need plenty of fasteners and printed parts, we will also need heat set inserts so be ready to get those in place too.
The printed parts list is in the X axis section, and the files start all with “*X_*” so you will need all of those.

X axis tensioner

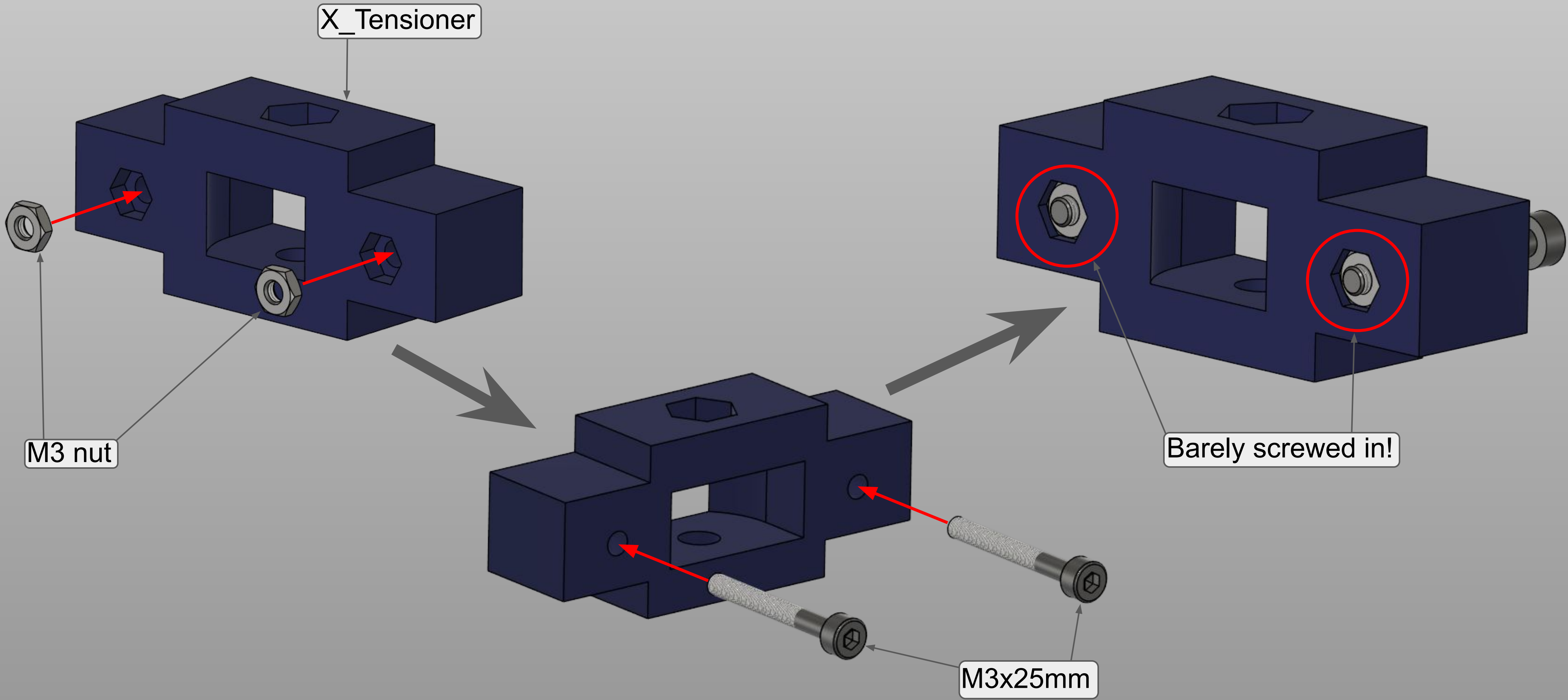


X axis tensioner

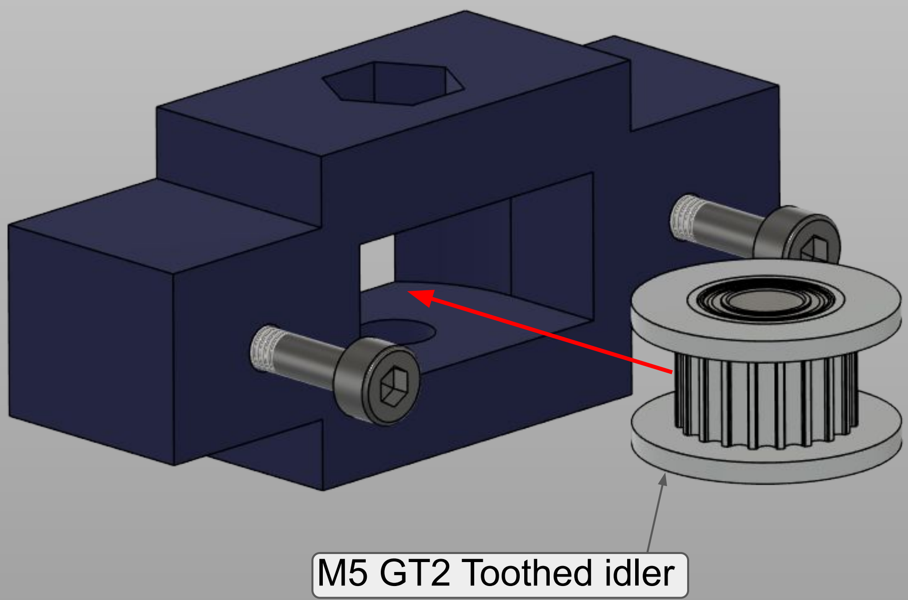


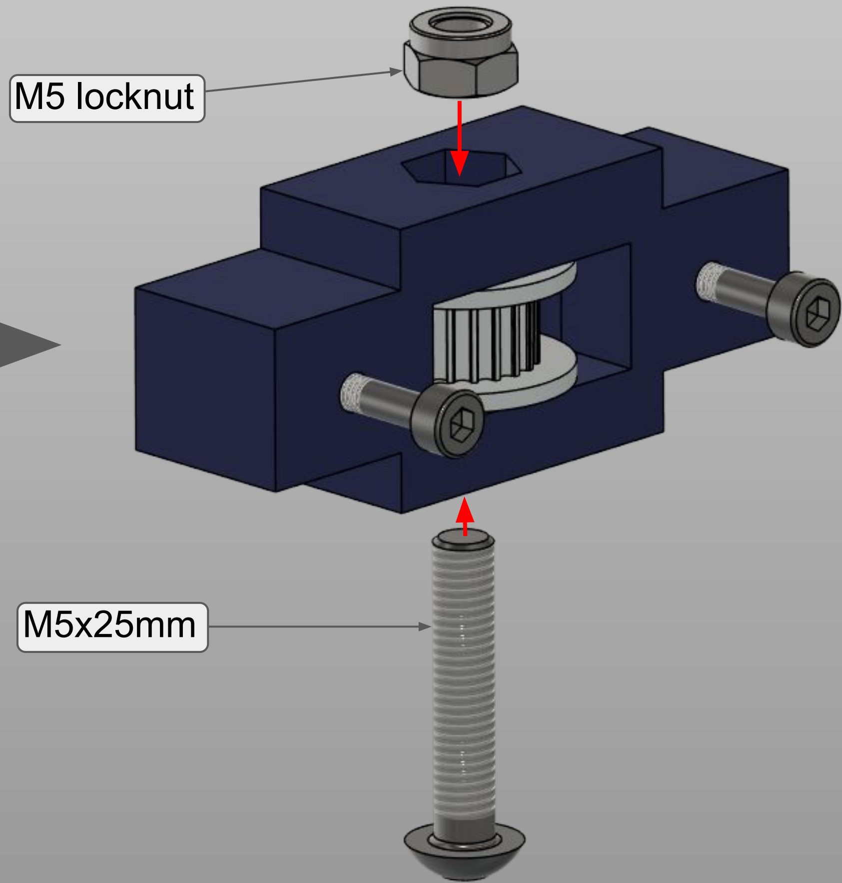
When adding the nuts, make sure to NOT tighten them but leave them as shown. It will be needed for sliding in the profile later on.

X axis tensioner



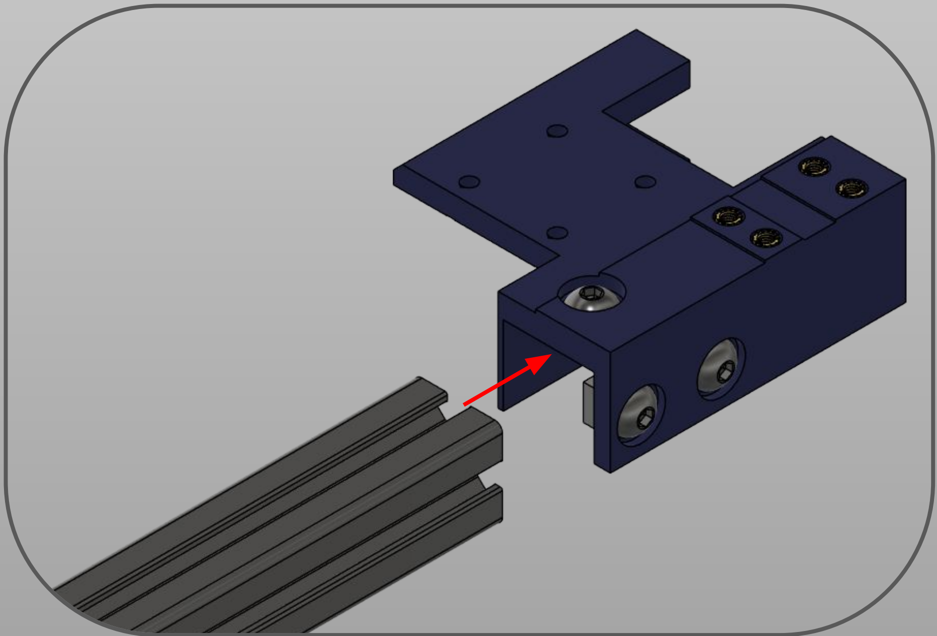
X axis tensioner





Note: *Make sure the idler spins freely, if not loosen the screw a bit until it does.*

X axis tensioner

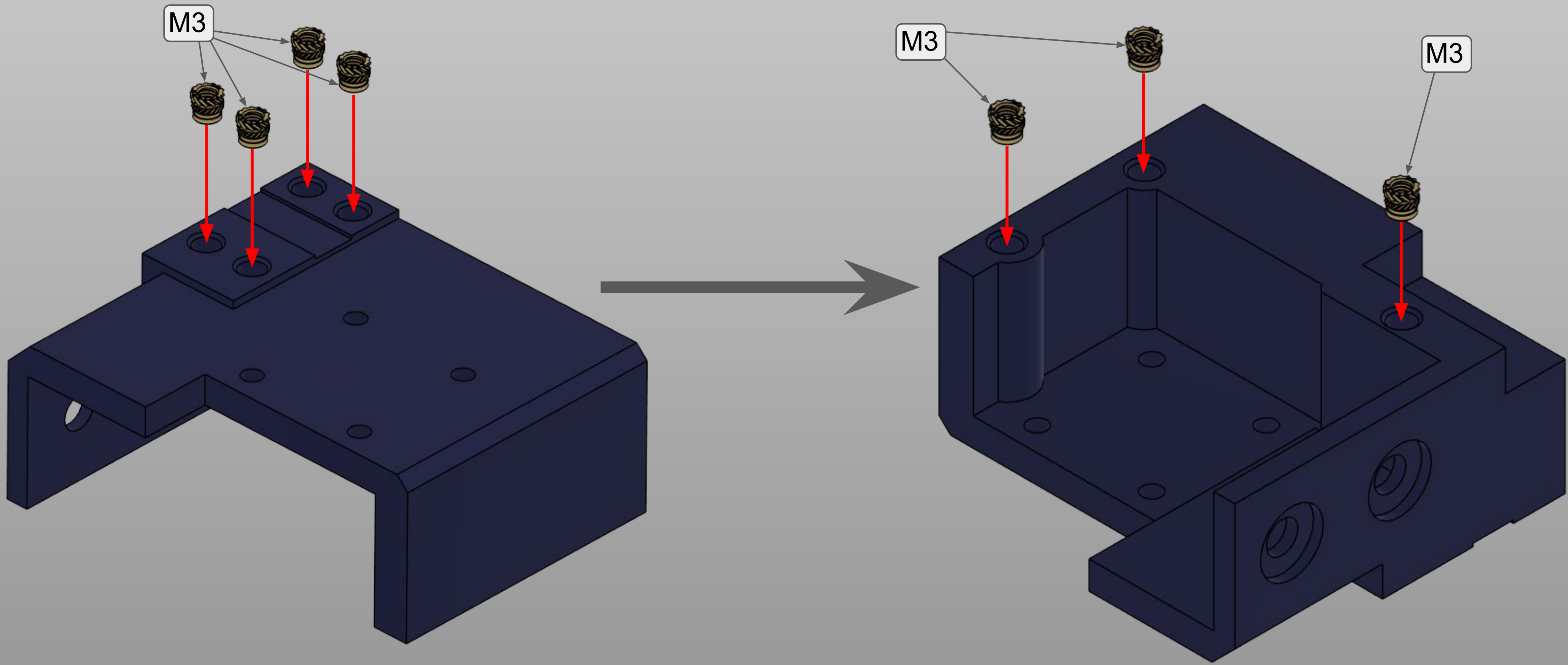


Slide in the 478mm profile as shown until it stops. Then tighten the three screws fully to secure it to the X axis beam.

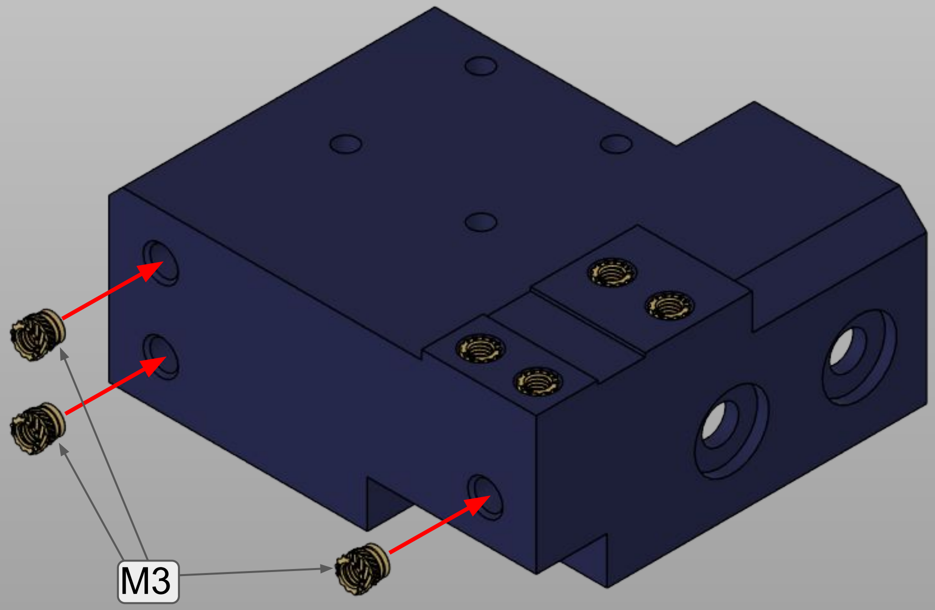
Then prepare the X rail as the Y before as shown below.

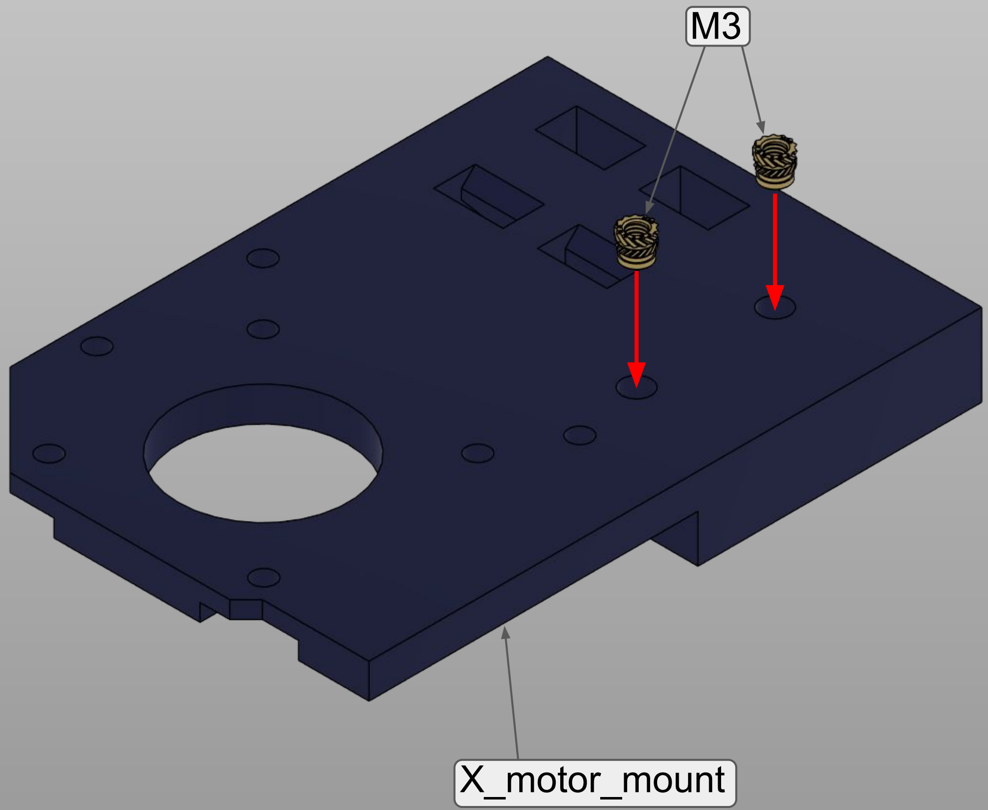
X axis

X motor mount and XY joint

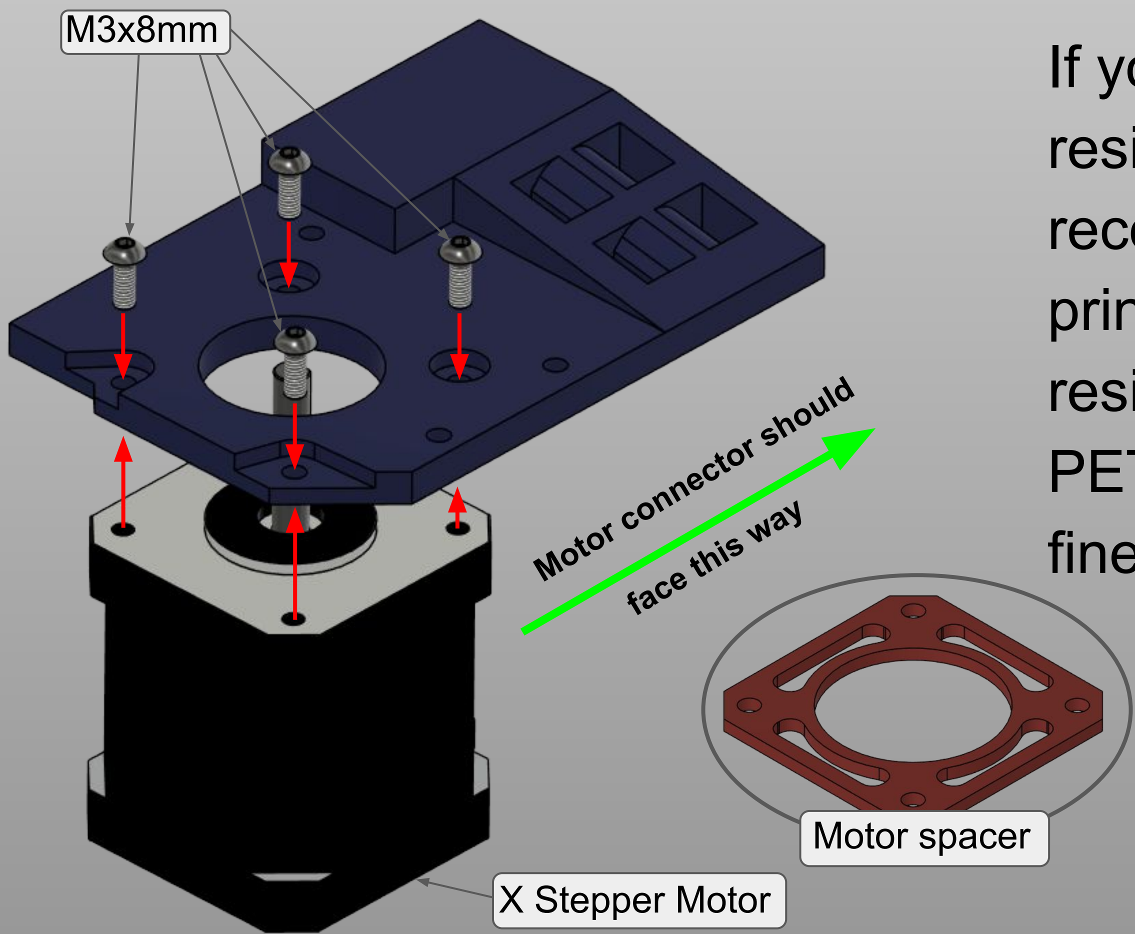


X motor mount and XY joint





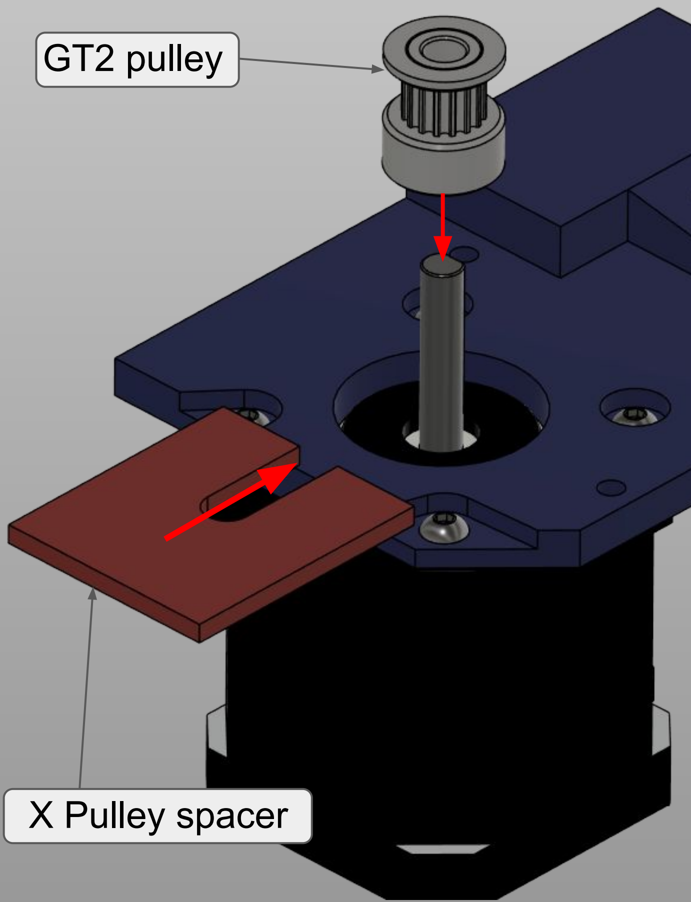
X motor mount and XY joint



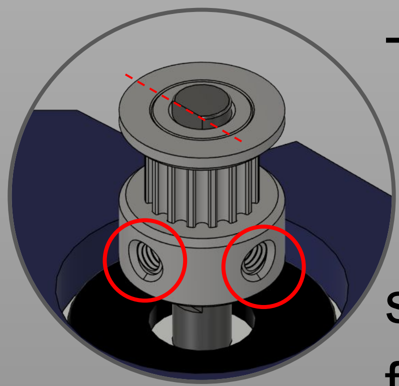
If you printed the pieces in non heat resistant materials, we strongly recommend to use M3x10mm screws and print the X_Motor_Spacer file in heat resistant materials like ASA, ABS or PETG (not as resistant, but should be fine).

Make sure the motor connector faces the correct way!

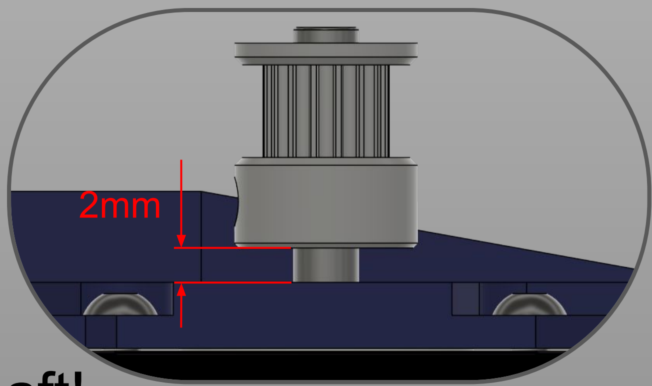
X motor mount and XY joint



Now get the GT2 pulley, undo the set screws enough to slide it in and use the pulley spacer as shown to get the right distance from the motor mount plate. The distance from the bottom of the pulley to the surface of the motor mount should be 2mm.

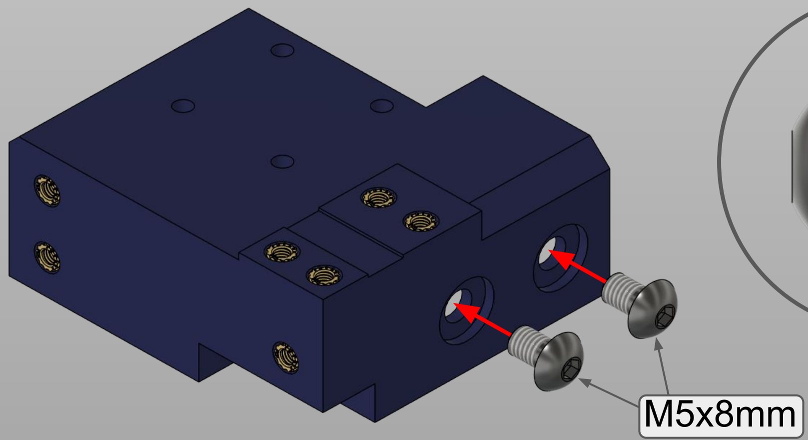


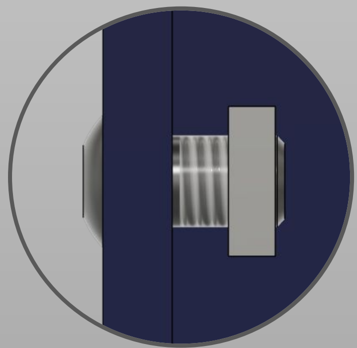
Then fix the pulley to the motor shaft as shown. Mind the flat side of the shaft!

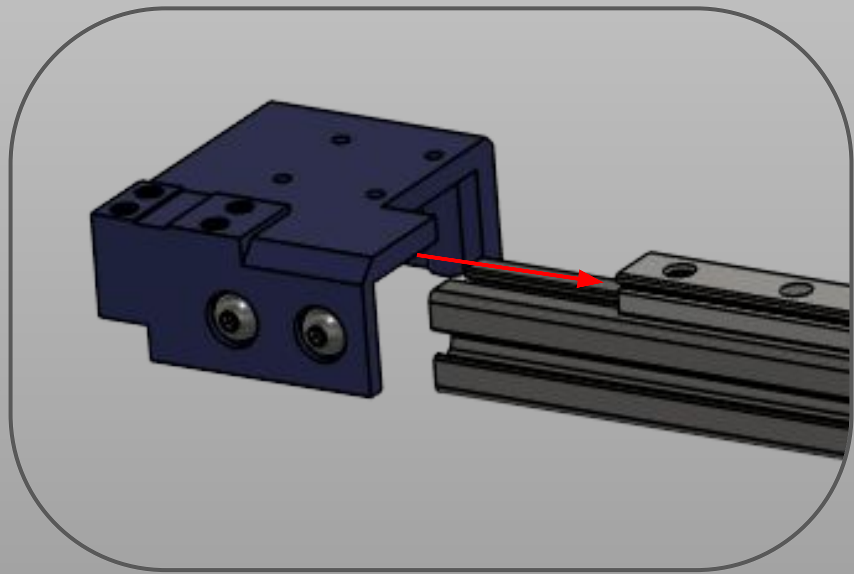


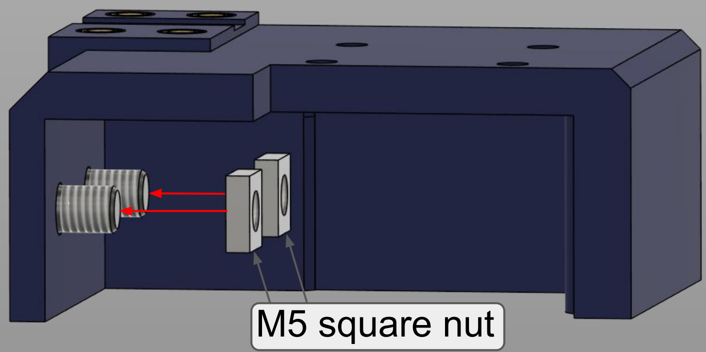
X motor mount and XY joint

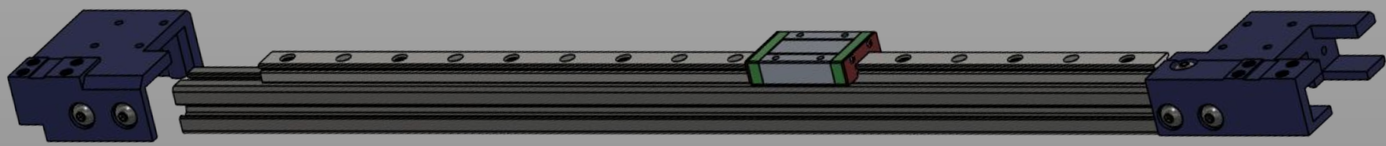
For mounting on the X beam, follow these steps. Do not tighten it yet! We need to align it before!







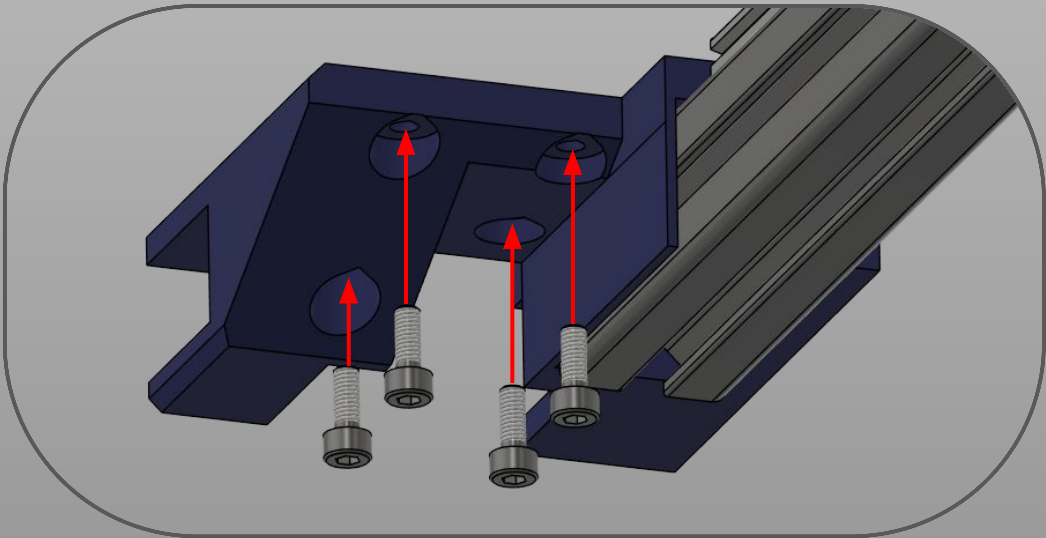


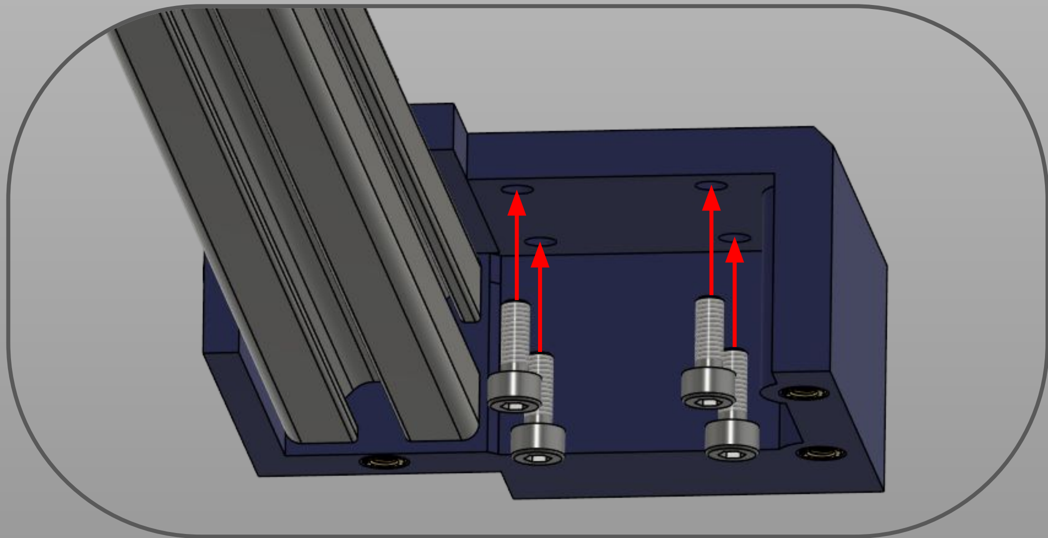


Mounting X axis on the Y rails

These next steps can be tricky and from them depends the squareness of the XY plane. To make sure the plane is as square as possible we will first keep all screws loosely in, then we will align the pieces and screw everything down.

For these steps we need 8 M3x8mm screws fitted:

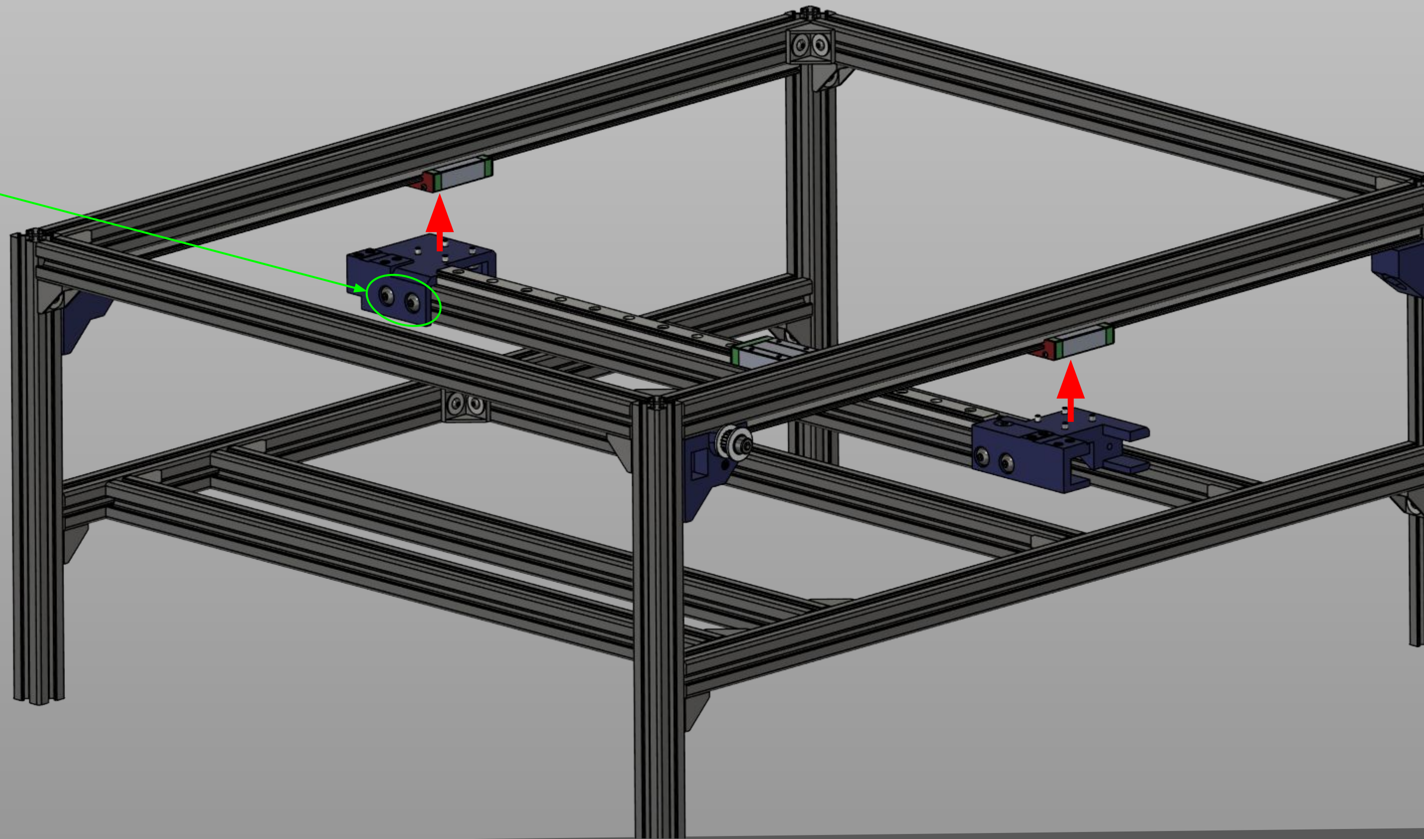




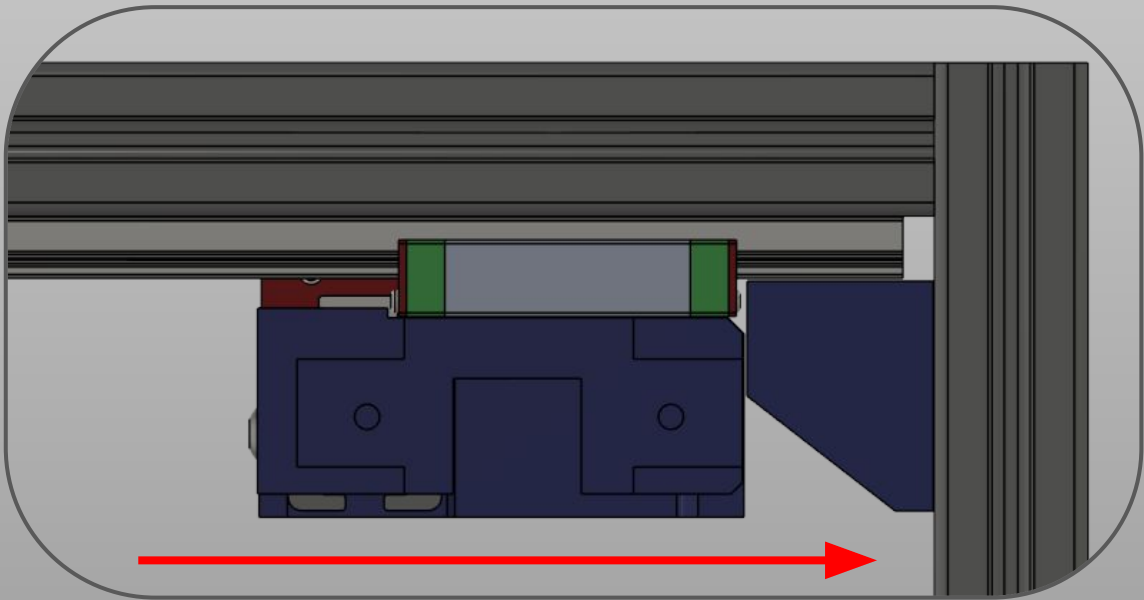
Mounting X axis on the Y rails

Align the X beam carefully and screw in **loosely** the printed parts to the carriages. Check **these** screws, they should be loose.

Pay close attention to the direction pieces are facing, they must be facing toward the front!!! It is crucial for the correct operation of the machine!



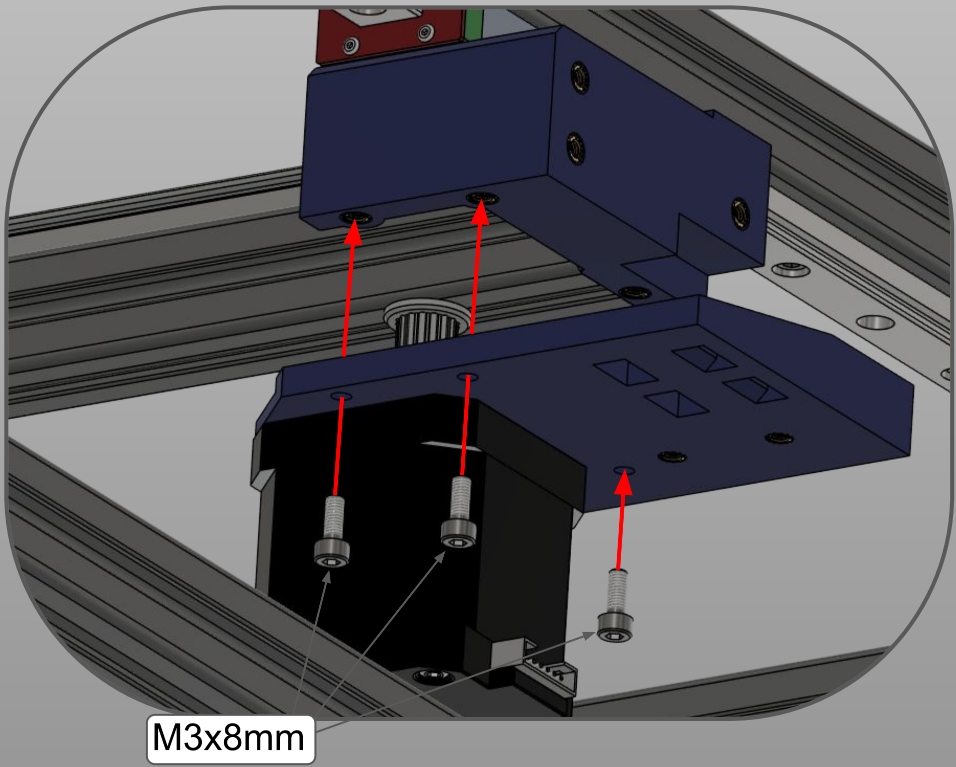
Aligning the axes

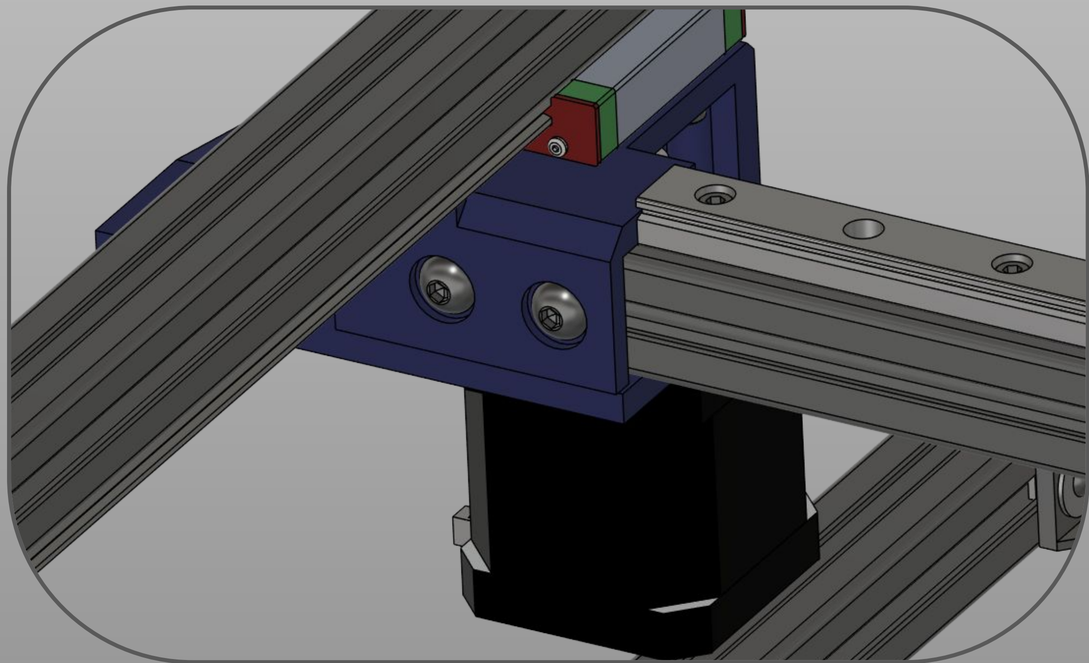


With the screws from the previous step loose, push the X axis so both carriages are touching the Y stoppers. Then, keeping them in this way, tighten all screws. Move back and forth the X axis to check if there are binding spots or unevenness. If the frame is square there should be none. Now let's finish the X axis.

X motor mount and XY joint

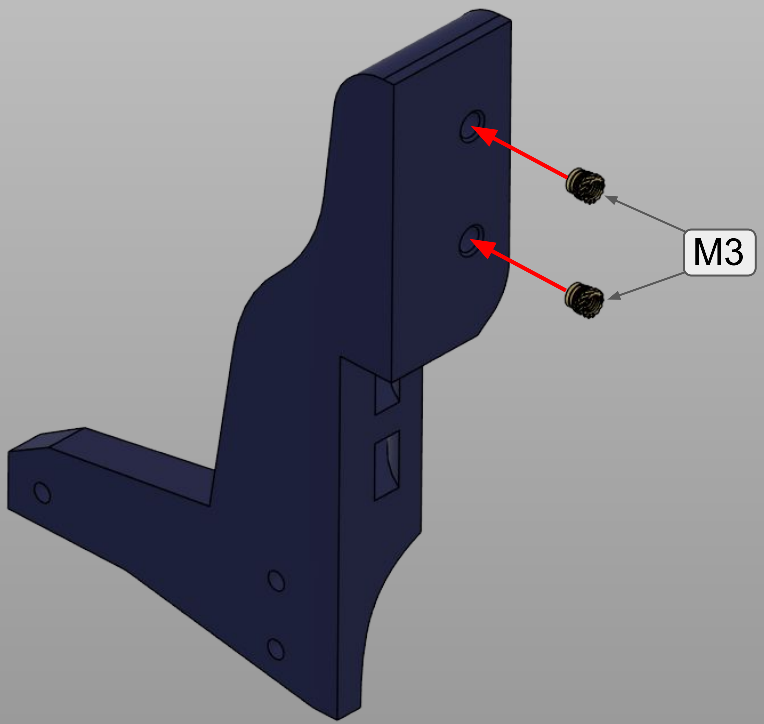
Fit the motor assembly as shown below, tighten all the screws and double check all the connections.

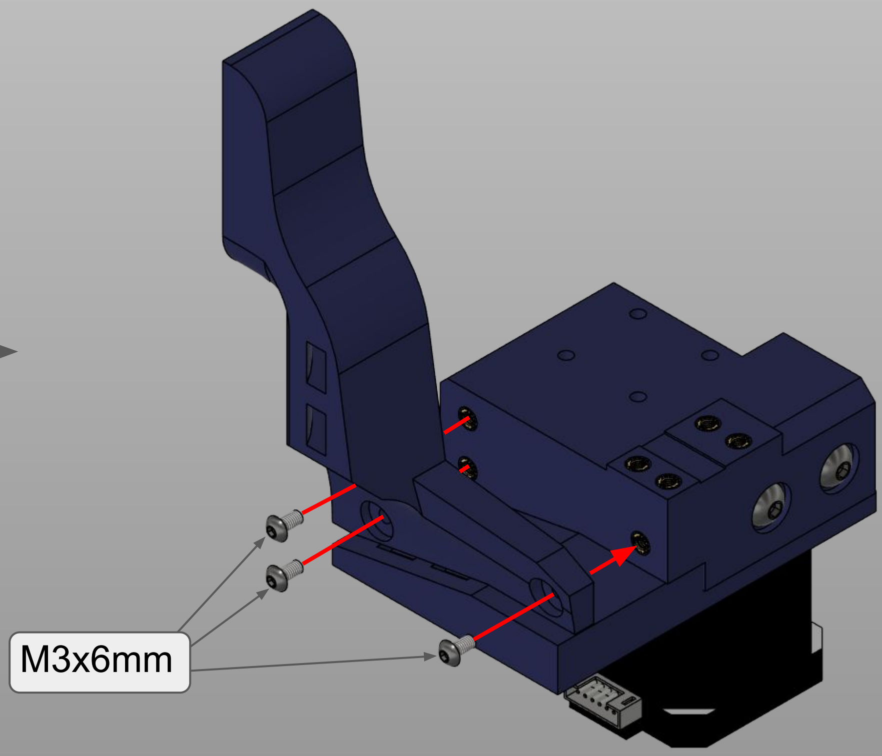




XY joint chain mount

The steps in these pictures can be completed before attaching the X axis, but this is a better moment to do it as they are bulky pieces



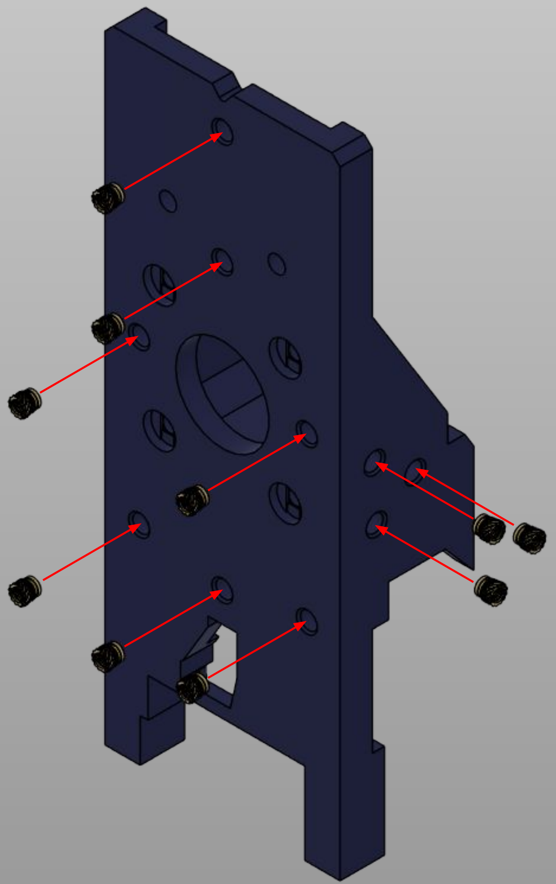


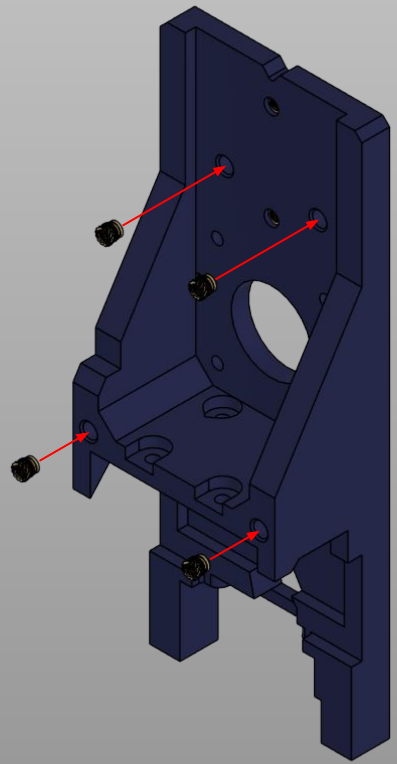
A word about naming

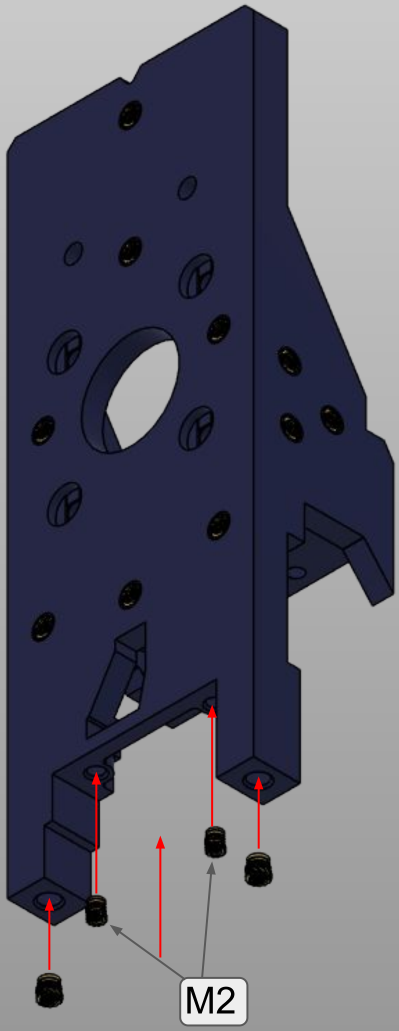
This section is called toolhead, because i come from the 3D printing world and this is the most familiar way to talk about it.
I just want to let you understand that with “toolhead” i mean the whole assembly where the two heads will be mounted, their linear rails and the Z motor.
This section is also relying on a “toolboard” PCB, a glorified connector board that allows easy unplugging of the toolhead from the main harness when needed.
This allows you to make maintenance easier.
Modifications could be made to implement the same board on the XY joint if the request is loud enough.

Printed parts - Base plate

Note: due to the large amount of inserts, when not specified use **M3 inserts**.



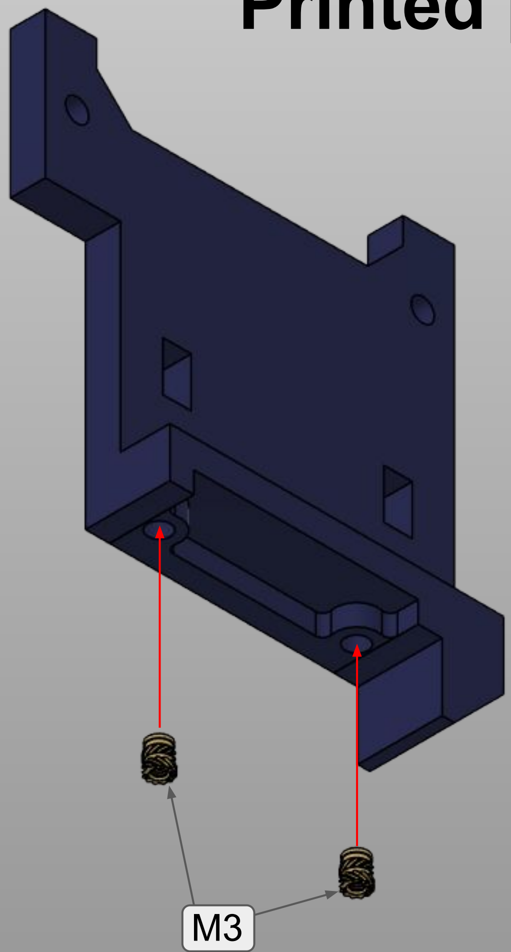


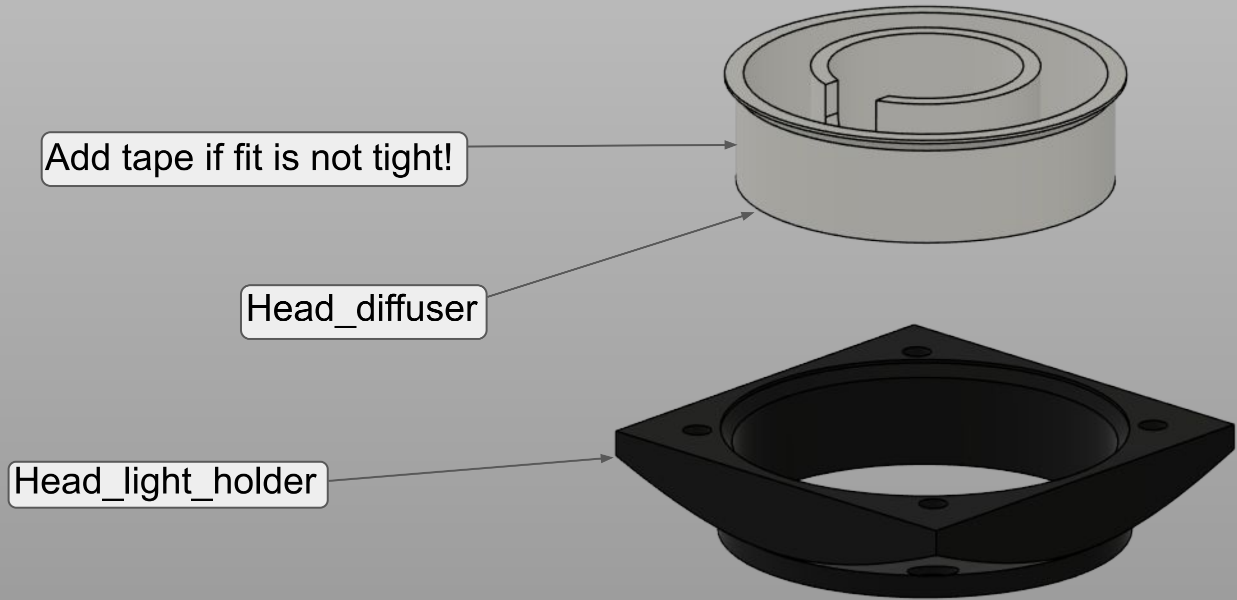


Toolhead

Printed parts - Rear cover and camera pieces

Concerning the camera parts, we will use only two screws right now, so keep this in mind.

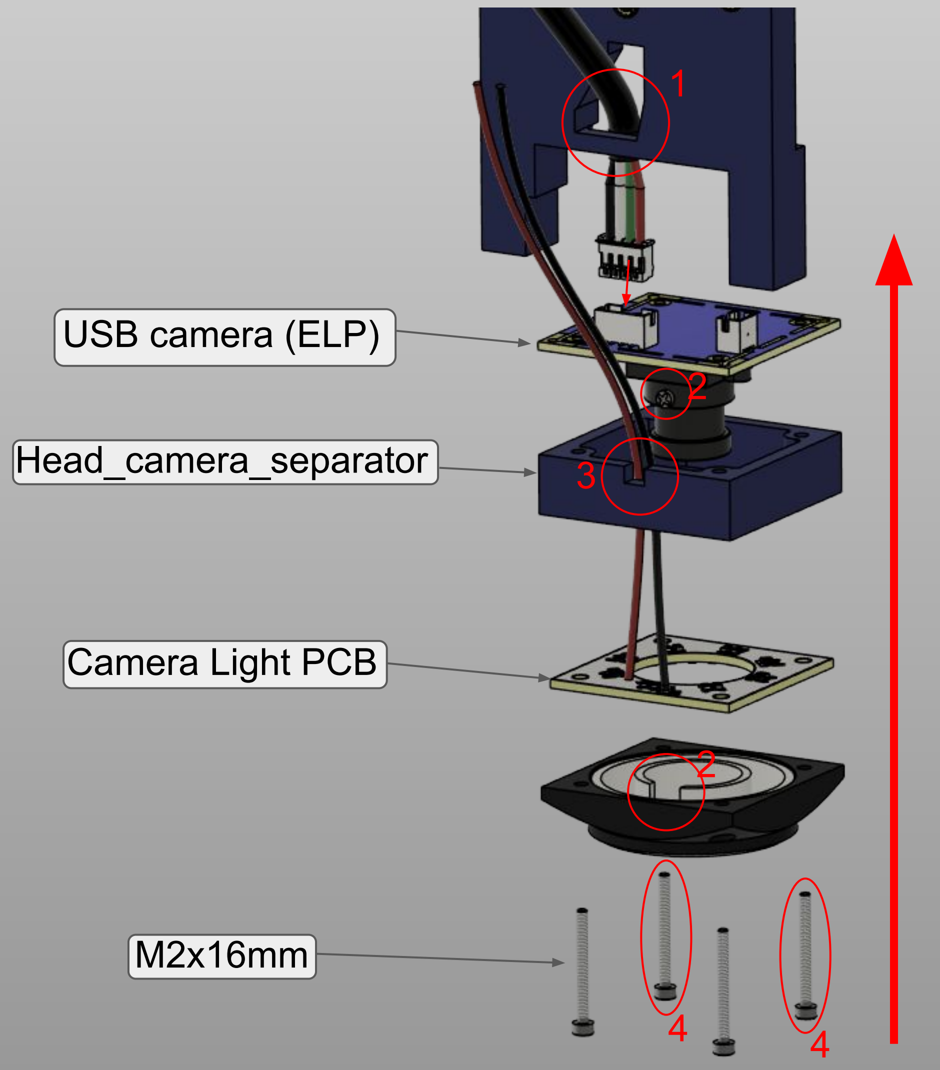


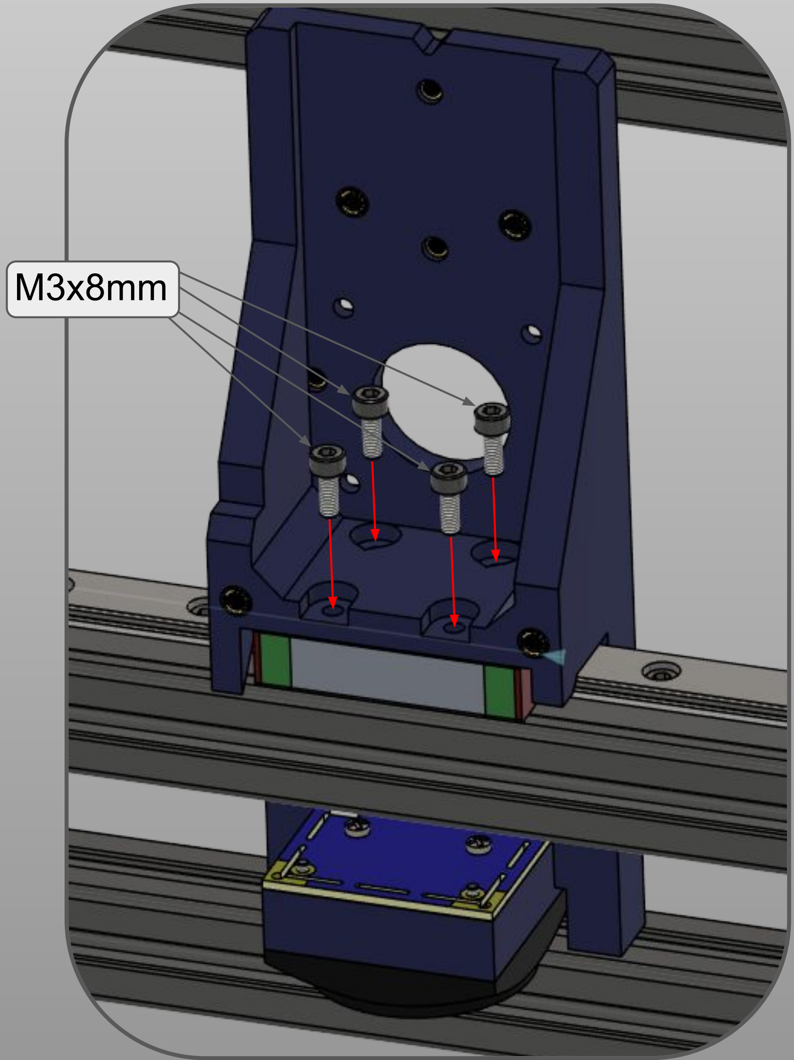


Head camera assembly

Assemble the camera section as shown, care must be taken for the following notes:

1) Cable from the camera MUST be installed prior and passed on this hole
2) Line up the camera alignment screw with the slot on the diffuser. We will take care of focusing later.
3) Make sure the wires from the lights are passing on this slot.
4) Slide in the screws anyway, we will tighten them later on but they keep everything aligned.





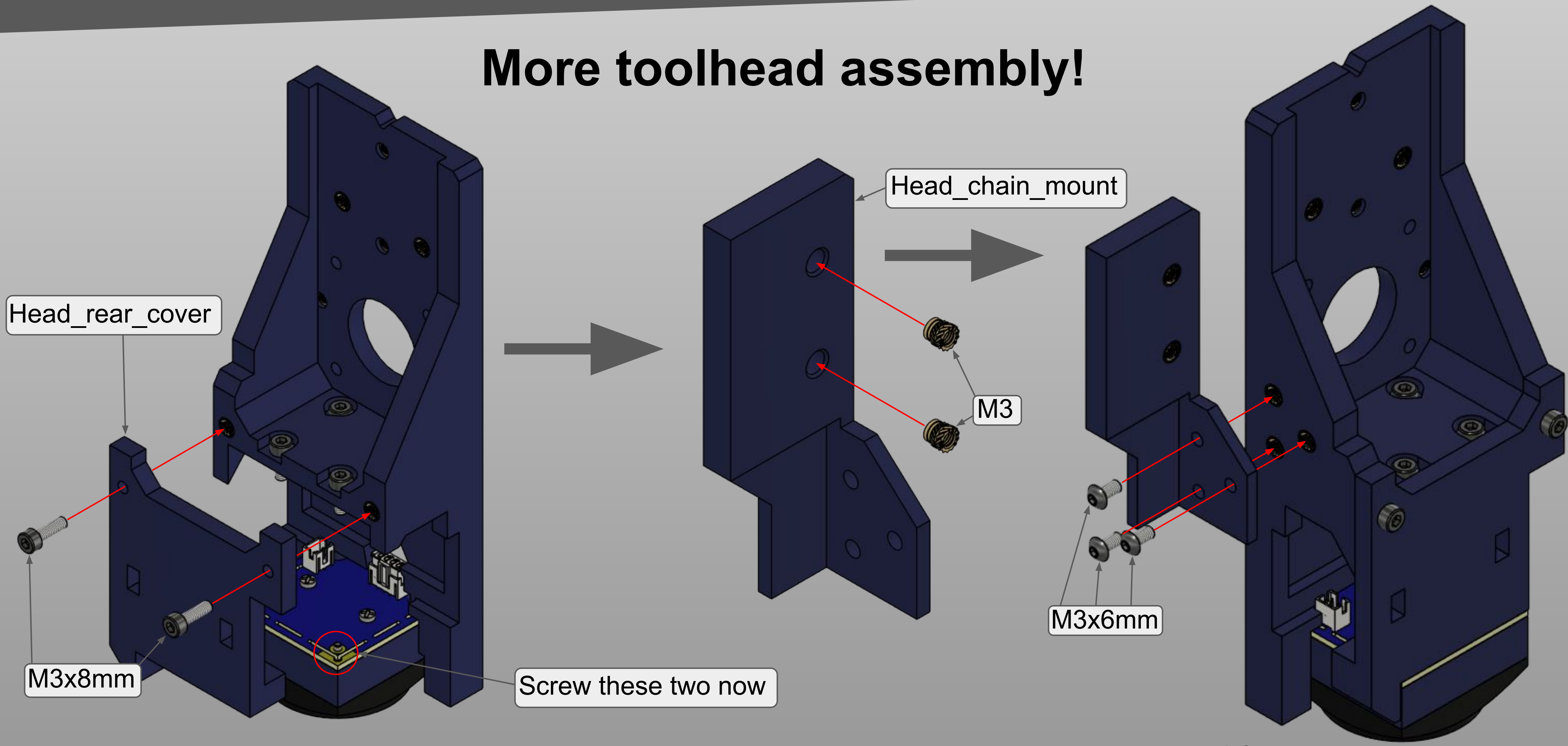
Mount the head to the X axis

Slide in carefully the toolhead on the X axis and fix it as shown. Tighten the screws since you will not get another chance to do it!
Pay attention to the direction, as the flat part MUST face the front!!!
Be especially careful to the camera's PCB, since it is exposed in this step.

Note: *From now on, until the cable management part is reached, cables will be shown ONLY when there needs to be particular precaution, as in the previous step.*

Toolhead

More toolhead assembly!



Note: *Assembly steps show ONLY the toolhead, but they need to be made after it is installed on the X axis!*

Toolhead

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